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A Chemical Reaction Similarity-Based Prediction Algorithm Identifies the Multiple Taxa Required to Catalyze an Entire Metabolic Pathway of Dietary Flavonoids

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Abstract

Flavonoids are polyphenolic phytochemicals abundant in plant-based, health-promoting foods. They are only partially absorbed in the small intestine, and gut microbiota plays a significant role in their metabolism. As flavonoids are not natural substrates of gut bacterial enzymes, reactions of flavonoid metabolism have been attributed to the ability of general classes of enzymes to metabolize non-natural substrates. To systematically characterize this promiscuous enzyme activity, we developed a prediction tool that is based on chemical reaction similarity. The tool takes a list of enzymes or organisms to match microbial enzymes with their non-native flavonoid substrates and orphan reactions. We successfully predicted the promiscuous activity of known flavonoid-metabolizing bacterial and plant enzymes.

Next, we used this tool to identify the multiple taxa required to catalyze an entire metabolic pathway of dietary flavonoids. Tilianin is a flavonoid-O-glycoside having biological and pharmacological activities, including neuroprotection. Using our prediction tool, we defined a novel bacterial pathway of tilianin metabolism that includes O-deglycosylation to acacetin, demethylation of acacetin to apigenin, and hydrogenation of apigenin to naringenin. We predicted and confirmed using in vitro experiments and LC-MS techniques that *Bifidobacterium longum* subsp. *animalis*, *Blautia coccoides* and *Flavonifractor plautii* can catalyze this pathway. Prospectively, the prediction-validation methodology developed in this work could be used to systematically characterize gut microbial metabolism of dietary flavonoids and other phytochemicals.

The bioactivities of flavonoids and their metabolic products can vary widely. We used an in vitro rat neuronal model to show that tilianin metabolites exhibit protective effect against H₂O₂ through reactive oxygen species (Delepine et al.) scavenging activity and thus, improve cell viability, while the parent compound, tilianin, was ineffective. These results are important to understand the gut microbiota-dependent physiological effects of dietary flavonoids.

eLife assessment

This paper reports **useful** findings regarding gut bacteria that metabolize dietary flavonoids, which can enhance, reduce, or otherwise alter the flavonoid bioactivities. With a newly developed bioinformatics tool, the authors predict bacterial species that can metabolize parts of the flavonoid tilianin. Formal proof of concept is missing, but if experimentally confirmed, the study will change the way we think about metabolism of flavonoids and would be of broad interest regarding gut bacterial metabolism. Most of the analyses are **compelling**, but others require further inquiry.

Introduction

Enzymes are exquisitely specific proteins that act as catalysts of biochemical reactions. They are evolved to recognize specific substrates and selectively catalyze reactions. However, some enzymes are capable of catalyzing additional reactions that may not be related to their native catalytic activities. This phenomenon is called enzyme promiscuity, which has been referred to as “underground metabolism” (Amin et al., 2019 [↗](#)), and includes the ability of enzymes to act on substrates other than which they evolved for (Khersonsky & Tawfik, 2010 [↗](#)). Promiscuity has a significant role in enzyme evolution and is potentially a useful tool for metabolic engineering. However, the promiscuous activity of an enzyme is not easy to detect using *in vivo* systems because of unknown reaction mechanisms, low levels of catalytic activities and lack of high-throughput methods to characterize enzyme-substrate pairs. Hence, promiscuous enzymes have not yet been comprehensively studied and remain to be fully utilized for metabolic engineering purposes. (Baas et al., 2013 [↗](#))

Flavonoids are a group of phytochemicals widely distributed in plants, fruits and vegetables that possess numerous bioactivities. They have a 15-carbon skeleton that consists of two phenyl rings (A- and B-rings) linked with a heterocyclic pyrane ring (C-ring). Depending on the substitutions on the central pyrane ring, flavonoids are classified into several subgroups: flavones, flavanones, flavonols, anthocyanins, flavan-3-ols and isoflavones. They are associated with various beneficial effects including anti-oxidant, anti-inflammatory and anti-cancer properties, making them attractive building blocks of nutraceutical, pharmaceutical and medicinal products. Their health promoting effects vary widely even among the same flavonoid subclass. (Gulshan et al., 2022 [↗](#))

The majority of flavonoids occur as glycosides in nature. Once flavonoids are ingested, they are partially absorbed in the small intestine, which makes them available to be metabolized by the gut microorganisms. Ingested flavonoids can be modified by removal of the sugar moiety, de/methylation and di/hydroxylation of A- and B-rings, or hydrogenation of the pyrane ring. Unlike plants, commensal gut microorganisms do not have specialized enzymes that utilize flavonoids as their native substrates. Hence, bacterial flavonoid metabolism is carried out through promiscuous enzyme activity. For example, methylation of flavonols is generally attributed to a flavonol O-methyltransferase (EC:2.1.1.76) enzyme (Schmidt et al., 2012 [↗](#)). According to the UniProt (UniProt, 2021 [↗](#)) and the KEGG (Kanehisa & Goto, 2000 [↗](#)) protein databases, this enzyme can only be detected in plant species. A recent study showed that an enzyme from *Streptomyces* sp. KCTC 0041BP that can catalyze the methylation of quercetin, luteolin, myricetin, quercetin 3-O- β -D-glucoside and fisetin (Darsandhari et al., 2018 [↗](#)) by cloning and heterologously expressing the enzyme in *E. coli*. We recently used *in silico* predictions and *in vitro* experiments to show that a chalcone synthase-like bacterial polyketide synthase can catalyze the cleavage of naringenin’s C-ring (Gülşan et al., 2022 [↗](#)). In plants, flavanone ring cleavage is catalyzed by chalcone isomerases (Braune et al., 2016 [↗](#)), which suggests that the polyketide synthase’s ability to cleave the C-ring of

naringenin reflects a promiscuous activity. Despite the growing interest in dietary flavonoids owing to scientific evidence regarding their health-promoting effects on the host, the enzymatic pathways and bacterial species responsible for the metabolism of these polyphenols in the intestine are largely unknown.

Various *in silico* approaches using rule- and/or machine learning-based methods have been developed to predict the metabolism of xenobiotics, e.g., PROXIMAL (Yousofshahi et al., 2015 [↗](#)), enviPath (Wicker et al., 2016 [↗](#)), Meteor Nexus (Judson et al., 2015 [↗](#)), and BioTransformer (Wishart et al., 2022 [↗](#)). Other approaches in the field utilize reinforcement learning methods (Yousofshahi et al., 2011 [↗](#)), (Delepine et al., 2018 [↗](#)), (Koch et al., 2020 [↗](#)), (Zheng et al., 2022 [↗](#)) to identify novel synthesis pathways in host organisms to produce natural products. However, there have been limited attempts to construct prediction tools for gut microbial metabolism of dietary flavonoids and to investigate the role of enzyme promiscuity. In the present study, we develop a chemical reaction class similarity-based prediction tool for investigating microbial flavonoid metabolism. We analyzed the KEGG database to identify and tabulate Reaction Classes (RClass) that represent the known flavonoid metabolism reactions by plant enzymes. The user can provide an enzyme, a bacterial species or a group of species as inputs. The tool generates predictions for the input's potential to metabolize dietary flavonoids by comparing the RClasses that are associated with the input enzyme(s) and the key RClasses that are chosen as representatives of gut microbial flavonoid metabolism reactions. Our approach in using RClasses instead of individual reactions allows the tool to introduce a degree of promiscuity instead of relying on matching the target molecule to the exact substrates of known reactions.

We used the above prediction tool to investigate if multiple species can collectively catalyze a multi-step pathway of flavonoid metabolism through promiscuous enzymatic action. We predicted the role of individual gut bacterial species in the degradation of tilianin via *O*-deglycosylation, demethylation and hydrogenation. Using *in vitro* culture experiments, we validated our predictions that *Bifidobacterium longum subsp. animalis*, *Blautia coccoides* and *Flavonifractor plautii* each catalyze particular steps in the pathway and together perform the degradation of tilianin to naringenin. Further, we demonstrated that tilianin and its metabolites have different antioxidant activities in a rat neuronal cell model. Prospectively, the prediction-validation methodology developed in this work could be used to systematically characterize the metabolism of flavonoid metabolites and identify the responsible enzymes and species, directly linking the formation of bioactive flavonoid metabolites to abundance of specific gut bacteria.

Results

RClass similarity successfully predicts the activity of known flavonoid-metabolizing plant and bacterial enzymes

To match a given enzyme with its non-native polyphenolic substrates and promiscuous flavonoid metabolism reactions, the tool retrieves the RClasses that are associated with the enzyme(s) from the KEGG database, then finds whether those RClasses are similar to the key RClasses that are chosen as representatives of flavonoid metabolism reactions as detailed in the Methods section. These key flavonoid reactions are *O*-deglycosylation, C-hydrolysis, de/hydrogenation, de/methylation, di/hydroxylation, ring cleavage, anthocyanin/anthocyanidin di/hydroxylation and anthocyanin/anthocyanidin de/hydrogenation reactions. **Figure 1** [↗](#) shows the general workflow for the predictions. The output contains the reaction type, similarity score and potential flavonoid substrates.

We evaluated the tool's accuracy by using known flavonoid-metabolizing enzymes from plants as positive controls, and glycolysis enzymes that are not expected to utilize flavonoids as negative controls (**Table 1** [↗](#)). The enzymes and their catalytic activities were collected from the BRENDA

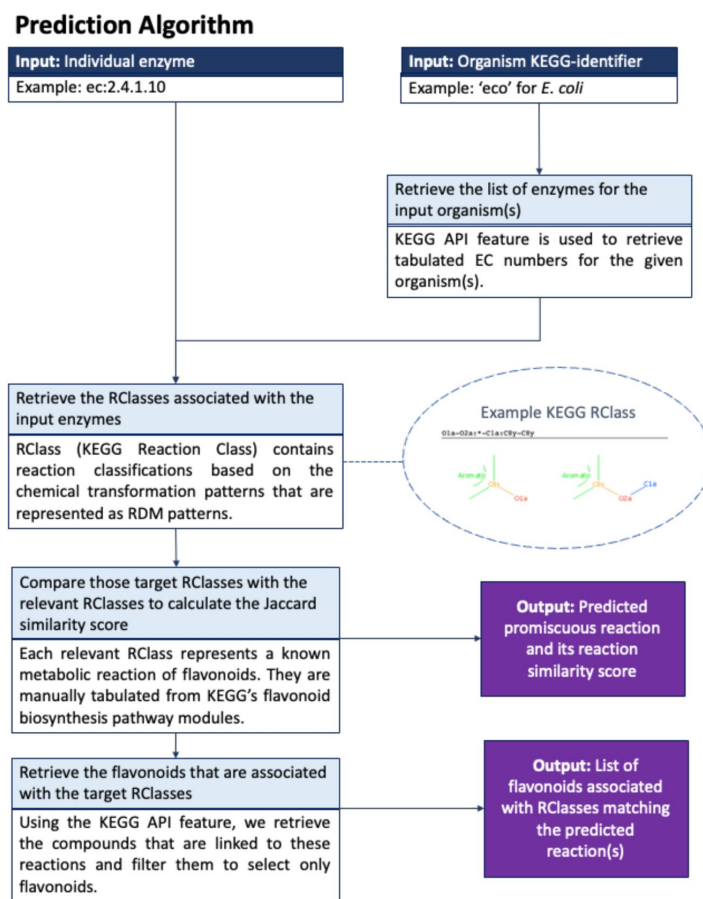


Figure 1

Workflow the algorithm utilizes to generate predictions for a given enzyme or an organism

enzyme database (Schomburg et al., 2004 [↗](#)). Positive control enzymes were chosen to test each reaction type that the tool is expected to predict. We could not test the di/hydroxylation of anthocyanin/anthocyanidin as BRENDA does not catalog any enzymes that are known to carry out this reaction. Negative control enzymes were chosen from the Glycolysis/Gluconeogenesis Pathway in the KEGG database to include at least one enzyme from each Enzyme Commission hierarchical group. For the test runs, we included enzymes that catalyze reactions utilizing molecular oxygen, because plant metabolism is not restricted to anaerobic environments.

Our analysis shows that Rclass-based similarity can predict the correct reactions for known flavonoid-metabolizing enzymes for ring cleavage, di/hydroxylation, de/methylation, O-deglycosylation, C-hydrolysis, de/hydrogenation and anthocyanin/anthocyanidin de/hydrogenation reactions. We observed that RClass similarity scores for the correctly predicted reactions are above 0.8. Therefore, we set this number as the cutoff value for predictions. Chalcone synthase is classified a transferase enzyme, but we previously found that it can catalyze the C-ring cleavage of naringenin through promiscuous enzymatic action (Gülşan et al., 2022 [↗](#)). This was correctly predicted by the tool developed in this work (Table 1 [↗](#)). This indicates that our algorithm can also be used to identify enzymes that act on flavonoids to carry out specific metabolism through promiscuous activity. Importantly, our tool did not predict any flavonoid-metabolizing reactions for the glycolysis enzymes as expected.

Enzyme sequence similarity does not necessarily indicate reaction class similarity

Enzymes having similar protein sequences have been shown to often perform similar catalytic functions. Hence, sequence alignment approaches have been used to classify enzymatic functions. The tool developed in this work instead analyzes functional similarity between enzymes by comparing the enzymes' reaction classes. To determine if sequence and reaction class similarity are correlated for enzymes involved in O-deglycosylation, C-hydrolysis, de/hydrogenation, di/hydroxylation, and de/methylation, we compared the two metrics for these enzymes from five organisms: *E. coli*, *F. plautii*, mouse ear cress, mouse and human. (Table 2 [↗](#))

We found no significant correlation between sequence similarity and reaction class similarity in all but one of the aforementioned comparisons. The only exceptions are within the comparison of the enzymes for O-deglycosylation in mouse and O-hydrolysis in *E. coli*. With a R-value of 0.327, and a p-value of 0.011, there is a weak, albeit significant, correlation in mouse O-deglycosylation findings. Similarly, the correlation for O-hydrolysis by *E. coli* enzymes show a significant correlation value of 0.567. There are three comparisons (di/hydroxylation in *F. plautii*, de/hydrogenation in *E. coli*, and de/hydrogenation in human) for which neither an R-value nor p-value could be calculated. In all other comparisons, the correlations were insignificant. Four illustrative sample comparisons are shown in Figure 2 [↗](#) for di/hydroxylation in mouse ear cress, de/hydrogenation in human, O-deglycosylation in *F. plautii*, and di/hydroxylation in *E. coli*. The correlation is weak in all four comparisons, with R-values of 0.031, 0.086, 0.185, and 0.333, respectively. Furthermore, these correlations are not statistically significant, with p-values of 0.842, 0.711, 0.766, and 0.244, respectively. The result of these comparisons shows that sequence similarity does not necessarily indicate reaction class similarity.

Mono- and co-culture experiments confirm tilianin metabolism by selected multiple gut bacteria

We next utilized the prediction tool to investigate the multiple gut bacterial reactions that are required to metabolize a flavonoid glycoside, tilianin, found in many medicinal plants, into an aglycone derivative. Using the tool developed in this work, we are proposing a metabolic pathway that proceeds through O-deglycosylation of tilianin to acacetin, then demethylation of acacetin to

	Input Enzyme ID	Enzyme Description	Catalytic Activity	Predicted Reactions	Similarity / Likelihood Score
Positive Control	ec:2.3.1.74	Chalcone synthase	3 malonyl-CoA + 4-coumaroyl-CoA = 4 CoA + naringenin chalcone + 3 CO ₂	Ring cleavage	1
Positive Control	ec:1.14.14.81	Flavonoid 3',5'-hydroxylase	a flavanone + [reduced NADPH-hemoprotein reductase] + O ₂ = a 3'-hydroxyflavanone + [oxidized NADPH-hemoprotein reductase] + H ₂ O	Di/hydroxylation	1
				Ring cleavage	0.75
				O-Hydrolysis	0.7
Positive Control	ec:2.1.1.232	Naringenin 7-O-methyltransferase	S-adenosyl-L-methionine + (2S)-naringenin = S-adenosyl-L-homocysteine + (2S)-sakuranetin	De/methylation	1
				O-Hydrolysis	0.8
Positive Control	ec:2.1.1.75	Apigenin 4'-O-methyltransferase	S-adenosyl-L-methionine + apigenin = S-adenosyl-L-homocysteine + acetin	De/methylation	1
				O-Hydrolysis	0.8
Positive Control	ec:2.4.1.91	Flavonol 3-O-glucosyltransferase	Flavonol + UDP-alpha-D-glucose = a flavonol 3-O-beta-D-glucoside + H(+) + UDP	O-Hydrolysis	1
				De/methylation	0.8
				C-Hydrolysis	0.6
Positive Control	ec:3.7.1.4	Phloretin hydrolase	H ₂ O + phloretin = 1,3,5-trihydroxybenzene + H(+) + phloretate	C-Hydrolysis	0.8
				Di/hydroxylation	0.7
				Ring cleavage	0.5
Positive Control	ec:5.5.1.6	Chalcone isomerase	A chalcone = a flavanone	Ring cleavage	1
Positive Control	ec:2.4.1.159	Flavonol-3-O-glucoside L-rhamnosyltransferase	Flavonol 3-O-beta-D-glucoside + UDP-beta-L-rhamnose = a flavonol 3-O-[alpha-L-rhamnosyl-(1->6)-beta-D-glucoside] + H(+) + UDP	O-Hydrolysis	0.8
				De/methylation	0.6
Positive Control	ec:2.1.1.42	Flavone 3'-O-methyltransferase	3'-hydroxyflavone + S-adenosyl-L-methionine = a 3'-methoxyflavone + H(+) + S-adenosyl-L-homocysteine	De/methylation	1
				O-Hydrolysis	0.6
Positive Control	ec:1.3.1.112	Anthocyanidin reductase [(2S)-flavan-3-ol-forming]	a (2S,3R)-flavan-3-ol + 2 NADP(+) = an anthocyanidin with a 3-hydroxy group + 2 NADPH + H(+)	Anthocyanin/din de/hydrogenation	1
				De/hydrogenation	0.73
Positive Control	ec:1.1.1.219	Dihydroflavonol 4-reductase	a (2R,3S,4S)-leucoanthocyanidin + NADP(+) = a (2R,3R)-dihydroflavonol + NADPH + H(+)	De/hydrogenation	1
				Di/hydroxylation	0.6
Negative Control	ec:5.4.2.2	Phosphoglucomutase	alpha-D-glucose 1-phosphate = D-glucose 6-phosphate	N/A	N/A
Negative Control	ec:4.2.1.11	Enolase	2-phospho-D-glycerate = phosphoenolpyruvate + H ₂ O	N/A	N/A
Negative Control	ec:6.2.1.1	Acetyl-CoA synthetase	ATP + acetate + CoA = AMP + diphosphate + acetyl-CoA	N/A	N/A
Negative Control	ec:2.7.1.1	Hexokinase	ATP + D-hexose = ADP + D-hexose 6-phosphate	N/A	N/A
Negative Control	ec:1.2.1.12	Glyceraldehyde-3-phosphate dehydrogenase	Acts on D-glyceraldehyde and some other aldehydes; thiols can replace phosphate	N/A	N/A
Negative Control	ec:3.1.3.80	2,3-bisphosphoglycerate 3-phosphatase	2,3-bisphospho-D-glycerate + H ₂ O = 2-phospho-D-glycerate + phosphate	N/A	N/A

Table 1.

Algorithm testing using positive and negative control examples. Positive controls were chosen among known flavonoid-metabolizing enzymes and negative control enzymes were chosen from glycolysis pathways as they are not expected to metabolize flavonoids.

Organism name	Reaction	R-value	p-value
Mouse Ear Cress	C-hydrolysis	0.698	0.054
Mouse Ear Cress	De/hydrogenation	0.533	0.276
Mouse Ear Cress	O-hydroysis	0.136	0.507
Mouse Ear Cress	Di/hydroxylation	0.031	0.842
Mouse Ear Cress	De/methylation	0.63	0.18
<i>F. plauti</i>	Di/hydroxylation	N/A	N/A
<i>F. plauti</i>	O-hydrolysis	0.185	0.766
<i>E. coli</i>	C-hydrolysis	0.423	0.478
<i>E. coli</i>	De/hydrogenation	N/A	N/A
<i>E. coli</i>	De/methylation	0.866	0.333
<i>E. coli</i>	Di/hydroxylation	0.333	0.244
<i>E. coli</i>	O-hydroysis	0.567	0.018
Human	Di/hydroxylation	0.563	0.245
Human	O-hydroysis	0.04	0.765
Human	De/hydrogenation	0.086	0.711
Mouse	De/hydrogenation	N/A	N/A
Mouse	Di/hydroxylation	0.526	0.362
Mouse	O-hydrolysis	0.327	0.011

Table 2.

Predicted reaction types for the tested organisms mouse ear cress, *F. plautii*, *E. coli*, human and mouse. Correlation values (R-value) between the sequence similarity and RClass similarity of predicted enzymes, and statistical significance of the correlation value (p-value) have been reported in the last two columns.

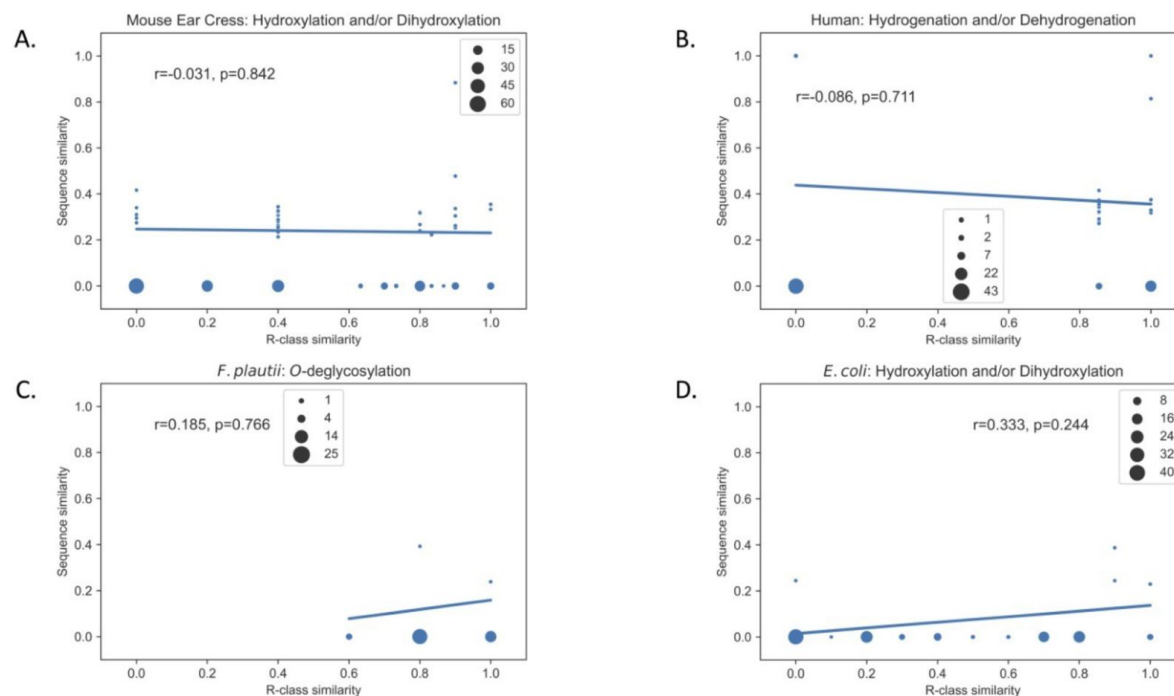


Figure 2

Comparison of sequence similarity and R-class similarity in predicted enzymes for A) di/hydroxylation in mouse ear cress, B) de/hydrogenation in human, C) O-deglycosylation of in *F. plautii* and D) di/hydroxylation in *E. coli*. The p-values and R-values are calculated in Python using the pearsonr function from the SciPy package.

apigenin, and finally hydrogenation of apigenin to naringenin (**Figure 3A** [↗](#)). Based on the RClass similarity based predictions, we identified *B. animalis* and *B. coccoides* as gut bacteria capable of the required *O*-deglycosylation and demethylation, respectively. A previous in vitro study identified an ene-reductase enzyme from *F. plautii* that can hydrogenate apigenin to naringenin (Schoefer et al., 2004 [↗](#)) (Schneider & Blaut, 2000 [↗](#)). Thus, we included this reaction in our proposed tilianin metabolism pathway.

To experimentally validate the predictions, we cultured selected species anaerobically in the presence of flavonoid substrate of interest for 24 hours, and analyzed the spent medium and cell pellets for accumulation of predicted metabolites. (**Figure 3B** [↗](#)) We observed that *B. animalis* can convert tilianin to acacetin by removing the glucose group attached to the flavone backbone as predicted. We also observed low concentrations of apigenin in the acacetin treated *B. animalis* culture, suggesting it can also catalyze demethylation on the B-ring of acacetin. While the absence of naringenin in the apigenin treated *B. animalis* culture indicates that it cannot hydrogenate apigenin to naringenin, we observed a substantial decrease in apigenin concentration suggesting its consumption by *B. animalis*. We searched for the monoisotopic masses of potential apigenin and naringenin microbial metabolites in tilianin treated *B. animalis* monoculture, which are di/hydroxylation and de/methylation products of apigenin and naringenin. We detected masses for hydroxylated apigenin and methylated naringenin (**Supplementary Figure S2** [↗](#)). The detected amounts of hydroxylated apigenin and methylated naringenin are statistically significant compared to the vehicle control. We also detected a compound whose mass matches with a dihydroxylated form of apigenin, although the accumulation of this compound was not statistically significant compared to vehicle control. These findings point to further, as yet uncharacterized flavonoid-metabolizing capabilities of *B. animalis*.

B. coccoides was predicted to remove the methyl group from acacetin to produce apigenin, but unable to metabolize tilianin and apigenin to acacetin and naringenin, respectively. The in vitro experiments and LC-MS assays confirmed these predictions. Similarly, *F. plautii*, a well-studied flavonoid-metabolizer strain, was not able to utilize tilianin and acacetin. When we incubated *F. plautii* with apigenin, the flavonoid substrate that is shown to be hydrogenated by *F. plautii* (Schoefer et al., 2004 [↗](#)) (Schneider & Blaut, 2000 [↗](#)), we did not detect any naringenin in the culture despite the decrease in apigenin concentration. However, we observed elevated amounts of 3-(4-hydroxyphenyl)propionic acid (34HPPA), a hydrolysis product of naringenin, in the apigenin treated samples compared to the vehicle control. The concentration difference is statically significant, which points to apigenin hydrogenation to naringenin and its further degradation to 34HPPA. Finally, our predictions also suggested that *E. coli* should not be able to perform any of the proposed tilianin metabolism reactions. Our findings indicate that *E. coli* indeed cannot utilize acacetin over a 24-hour incubation period. Surprisingly, it degraded both tilianin and apigenin in the culture. However, we did not observe any of the predicted metabolites in the *E. coli* culture, consistent with our predictions that *E. coli* cannot *O*-deglycosylate, demethylate and hydrogenate the selected compounds.

In order to rule out the possibility of chemical degradation of the compounds in the culture medium, we incubated the medium with 100 μ M tilianin, acacetin and apigenin and quantified the amount of substrate left in the culture after 24 hours. We did not observe any decrease in the compounds' concentrations. Further, the predicted degradation products of the selected parent compounds were not present in the culture medium. These results suggest that tilianin, acacetin and apigenin cannot be degraded to their *O*-deglycosylated, demethylated and hydrogenated metabolites in the absence of their metabolizer species.

O-deglycosylation of tilianin by *B. animalis* and demethylation of acacetin by *B. coccoides* are novel findings that were discovered by using the prediction-validation methodology developed in this work. We next investigated whether these two microorganisms can work together to metabolize tilianin all the way to apigenin. We incubated *B. animalis* and *B. coccoides* together and treated the

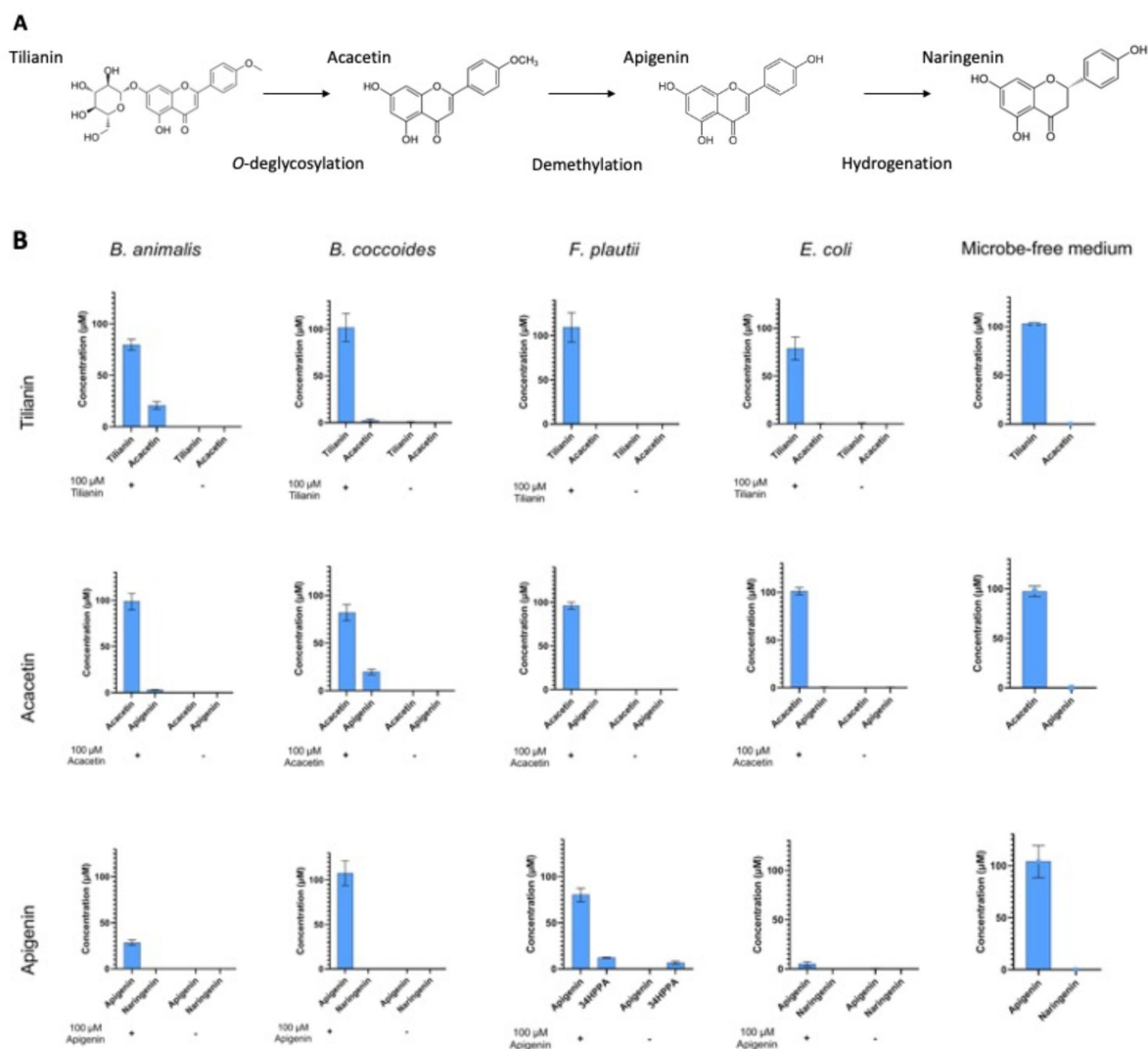


Figure 3

A) Proposed tilianein microbial metabolism pathway. B) Concentrations of parent compounds and their predicted immediate metabolites in the bacterial monocultures of *B. animalis*, *B. coocoides*, *F. plautii*, *E. coli* and microbe-free medium. Results have been represented in bar graphs and displayed together in a matrix format, where the columns represent the microbes, and the rows represent the parent compound that is spiked into the culture.

culture medium with 100 μ M tilianin for 24 and 48 hours. At the end of 24 hours, roughly 20% of tilianin was converted to acacetin. We observed low concentrations of apigenin, but the amount was not statistically significant compared to the vehicle control from the same timepoint. We observed an increase in acacetin concentration after 48 hours and a significant amount of apigenin formation. (Figure 4A) These results suggest that these two microorganisms can work together to catalyze tilianin *O*-deglycosylation to acacetin and acacetin demethylation to apigenin.

Among the selected bacteria, *B. coccoides* is the only strain that can metabolize acacetin over a 24-hour incubation period (Figure 3). Based on this finding, we hypothesized that this strain is essential for the complete degradation of tilianin. In the absence of *B. coccoides*, acacetin should not be metabolized to apigenin and should accumulate in the culture. We incubated *B. animalis* and *F. plautii* for 24 and 48 hours and treated the culture with 100 μ M tilianin. As expected, we did not observe any apigenin formation and acacetin concentration remained stable over time (Figure 4B). This finding indicates that each strain has a unique role in the proposed tilianin metabolism reactions and their unique contribution is essential to carry out the entire pathway.

We next evaluated our predictions against a computational tool that predicts biosynthetic pathways for natural products, BioNavi-NP (Zheng et al., 2022). The analysis was performed by providing tilianin as the query molecule. The tool suggested that the most plausible pathway for tilianin is its *O*-deglycosylation to acacetin, demethylation of acacetin to naringenin, hydrogenation of naringenin to apigenin, and C-ring cleavage to naringenin chalcone (Figure 5). While this pathway overlaps with the predictions from the RClass similarity-based tool developed in this work, BioNavi-NP does not associate these reactions with specific gut bacteria. The enzymes selected by BioNavi-NP are only expressed by plants, whereas our tool links the reactions with gut bacterial enzymes. Another alternative pathway predicted by BioNavi-NP for acacetin degradation to apigenin was through apigenin 7,4-dimethyl ether and genkwanin; however, this pathway is energetically costly and unlikely to occur naturally. BioNavi-NP also suggested that apigenin can be reduced and hydroxylated to 2-hydroxy-2,3-dihydrogenistein, tricetin or 2S-dihydrotricetin. These and other predictions are shown in Figure 5.

The stoichiometry of tilianin metabolism and conversion yields observed in the mono-culture experiments (Figure 3B) indicate that no other compounds should be formed to any significant extent. Using the monoisotopic masses of the compounds predicted by BioNavi-NP, we performed a mass-based search in our single culture experiments. None of the BioNavi-NP predicted compounds were detected in the cultures (data not shown).

Tilianin and its microbial metabolites elicit different biological activities

Rat pheochromocytoma cells (PC-12) originally derived from the adrenal gland have been extensively used as in vitro models in neurobiology owing to their ability to depict characteristics similar to cortical neurons (Wiatrak et al., 2020). PC-12 cells are especially desirable for early-stage protective screening against neurodegenerative diseases, which may be mimicked through disease dependent exogenous injury. In this study, we used PC-12 cells for an oxidative damage model as oxidative stress is a prevalent hallmark in several neurological disorders. We investigated the potential of tilianin and tilianin-derived metabolites to protect against hydrogen peroxide (H_2O_2) induced oxidative damage to PC-12 cells. Treatment of PC-12 cells with 200 μ M H_2O_2 sharply decreased cell viability by 67% with an associated increase in ROS level of 57% in comparison to the untreated control group (Figure 3-6). As expected, pretreatment with NAC significantly protected against loss of cell viability and elevation in ROS, with a 43% increase in viability and a 101% reduction in compared to the H_2O_2 treated group. Pretreatment with 10 μ M acacetin and 5 μ M naringenin significantly improved cell viability by 15% and 31% respectively. In contrast, tilianin and apigenin did not significantly impact viability. All tilianin pathway metabolites at 5 μ M significantly attenuated H_2O_2 -induced ROS generation with the greatest

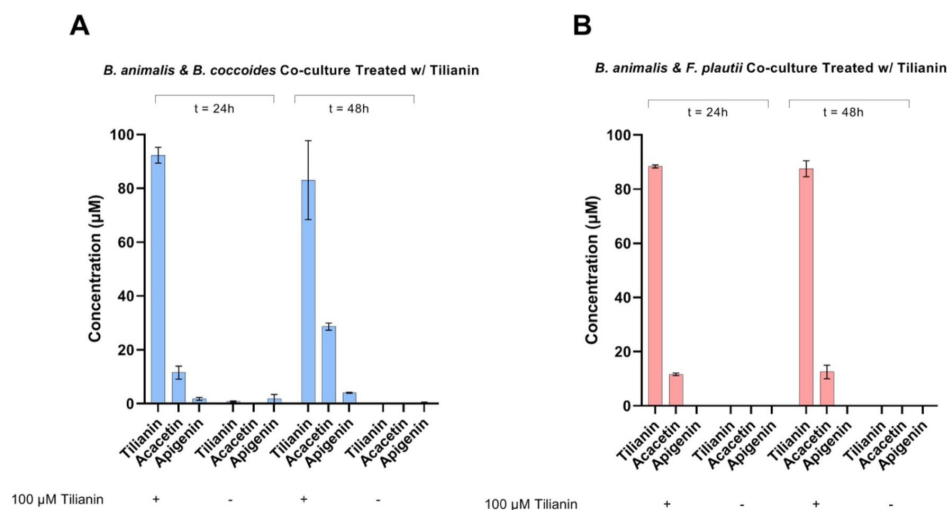


Figure 4

Concentrations of tilianin, acacetin and apigenin in the tilianin-treated co-culture of A) *B. animalis* and *B. coecoides*, and B) *B. animalis* & *F. plautii* after 24 and 48 hours of incubation.

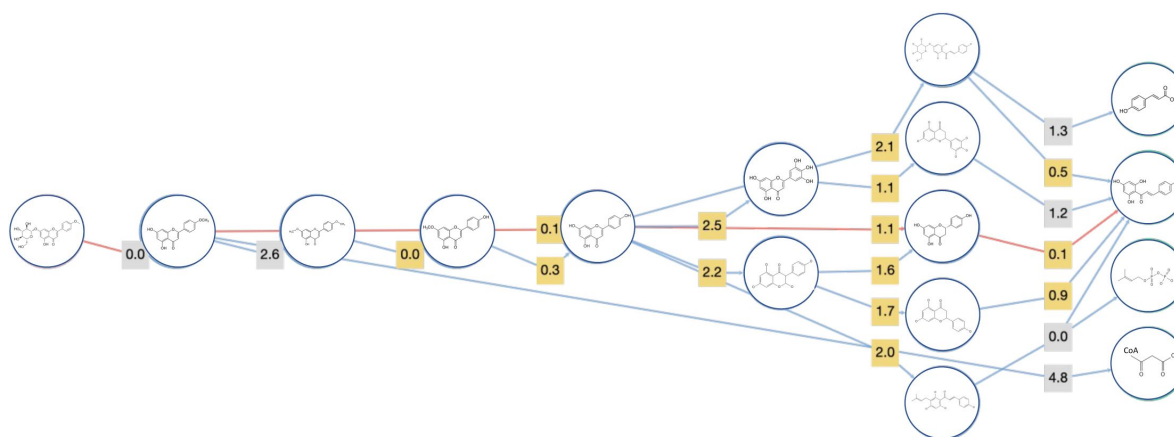


Figure 5

Tilianin metabolism pathway predicted by BioNavi-NP using the algorithm's default modes. The outputs were redrawn to provide higher resolution for the predicted compounds. The most plausible pathway is colored in red, which overlaps with the pathway proposed in this work. The calculated cost of each reaction step is reflected by the confidence score where lower score means higher reaction probability.

reduction of 90% achieved by acacetin. In contrast, tilianin itself failed to reduce ROS generation. Acacetin significantly attenuated ROS also at the higher concentration (10 μ M). Except for acacetin, pretreatment with tilianin or any other pathway metabolite at 10 μ M or higher concentration did not rescue cell viability or attenuate ROS generation (data not shown).

Discussion

Dietary flavonoids and the commensal gut microorganisms exert a two-way interaction; bacterial metabolism of flavonoids may alter flavonoid bioavailability. Previous studies have repeatedly shown that flavonoid metabolites may elicit different biological activities than their parent compounds, suggesting that gut bacterial species directly impact the profile of bioactive flavonoid metabolites. The most notable examples of microbiome-mediated flavonoid transformation that results in increase in biological activity can be seen in isoflavones (Morishita et al., 1989 [\[link\]](#)). Isoflavones are predominantly found in soy and their metabolism is directly linked to the microbiome variability. (Axelson & Setchell, 1981 [\[link\]](#)) Recent studies suggest that daidzein conversion to equol takes place in the intestine and is catalyzed by microbial reductase enzymes, however, the key gut microorganisms have not been fully characterized (Mayo et al., 2019 [\[link\]](#)). Equol shows stronger estrogenic (Setchell et al., 2005 [\[link\]](#)) (Jackson et al., 2011 [\[link\]](#)), anti-androgenic (Lund et al., 2004 [\[link\]](#)) and antioxidant (Wei et al., 2011 [\[link\]](#)) (Wei et al., 2011 [\[link\]](#)) activity than its food precursors daidzein and daidzin. Another possible metabolic conversion for daidzein is its C-ring cleavage to O-desmethylangolensin, which is likely produced by different bacterial taxa (Mayo et al., 2019 [\[link\]](#)). A majority of humans (80-90%) cannot produce equol as they metabolize a large fraction of daidzein to O-desmethylangolensin, which has no estrogenic activity. (Gardana et al., 2014 [\[link\]](#)) (Frankenfeld, 2011 [\[link\]](#)) These findings are crucial to understand the gut microbiota-dependent physiological effects of dietary flavonoids. Flavonoids may also alter the composition of microbiota profile by promoting beneficial bacteria as prebiotics, while inhibiting the growth of pathogenic species due to their anti-microbial properties. Studies have shown that flavonoids may stimulate the growth of beneficial bacteria *Lactobacillus spp.* and *Bifidobacterium spp.* in the intestine, while inhibiting the growth of pathogenic bacteria such as *Clostridium spp.* (Jiao et al., 2019 [\[link\]](#)) (Saad et al., 2016 [\[link\]](#)) (Ponzetto et al., 1988 [\[link\]](#)) (Anhe et al., 2015 [\[link\]](#)) (Carrera-Quintanar et al., 2018 [\[link\]](#)) Thus, flavonoids have the potential to therapeutically target the intestinal microbiome.

Similarly, increasing evidence points to a bidirectional interaction between the gut microbiota and the brain. Microbially mediated metabolites may interact with the central nervous system (CNS) indirectly through signaling pathways often involving the vagus nerve or directly by entering systemic circulation and permeating through the blood brain barrier (Bravo et al., 2011 [\[link\]](#)). Alternatively, the CNS impacts gut microbiome function through mediation of the gut environment. In a recent example, a brain injury mouse model noticed compositional changes in caecal microbial population dependent on the extent of brain injury, suggesting a causal link between brain damage and the gut microbiota population diversity (Houlden et al., 2016 [\[link\]](#)). As a result, the gut-brain-axis has been implicated as an important modulator of neurodegenerative disease pathology. For example, a study using an Alzheimer's disease - like pathology (ADLP^{APT}) transgenic mouse model noticed a reduction in neuropathological hallmarks namely, amyloid beta plaque formation, neurofibrillary tangles accumulation, and glial reactivity, with both frequent transfer and transplantation of gut microbiota from healthy mice (Kim et al., 2020 [\[link\]](#)). Similar clinical studies point to dysbiosis as a dysregulator of the microbial metabolite profile capable of interacting with the CNS (Vogt et al., 2017 [\[link\]](#)) (Cattaneo et al., 2017 [\[link\]](#)). Despite the growing consensus linking dysbiosis to neurodegenerative diseases, the driving cause of many neuronal diseases still remain unclear, causing difficulty in development of treatment targets. One hypothesis considers oxidative stress resulting from the over accumulation of reactive oxygen species as a significant contributor to neuronal damage. Dysregulated reactive oxygen species (ROS) negatively interact with cellular structures and disrupt critical pathways, which then leads

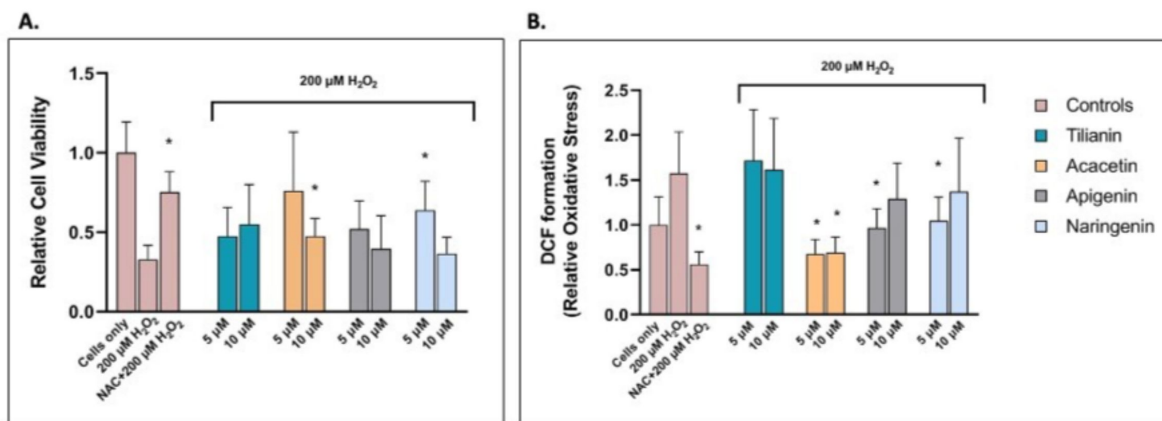


Figure 6

PC-12 response too 200 μ M H_2O_2 stress treatment for 4 hours following 1 hour pretreatment with tilianin, pathway metabolites (acacetin, apigenin, naringenin), or control conditions.

A. Viability was detected using calcein violet and shown as fraction of the untreated condition. B. The ROS level was assessed using DCFDA shown as a fraction of the untreated condition. N = 3 (figure shown is representative of all experiments); n = 4; *P < 0.05.

to further disease pathogenesis (Singh et al., 2019). Thus, this study seeks to determine the radical scavenging capacity of microbially mediated metabolites to elucidate the role of microbial flavonoid metabolism on oxidative stress protective potential.

In this work, we present a reaction class-similarity based prediction tool to identify multiple gut bacterial species that can together catalyze a multi-step pathway for flavonoid metabolism through promiscuous enzymatic activity. We predicted and experimentally validated that the sugar moiety attached to the A-ring of tilianin can be removed by *B. animalis* to produce acacetin, and that acacetin can be methylated to apigenin by *B. coccoides*. Recently, Yang et al. (Yang et al., 2021) reported hydrogenation of apigenin to naringenin by *F. plautii*. Based on these findings, we propose a novel gut microbial pathway for tilianin degradation in the intestine. Further, we used a rat neuronal cell model to show that tilianin metabolites can protect against oxidative stress induced loss in cell viability, whereas the parent compound tilianin was ineffective.

Gut microbial metabolism of dietary flavonoids, the largest class of dietary phenolic phytochemicals with over 6,000 structurally distinct compounds (Panche et al., 2016), has garnered wide interest from the scientific community due to the compounds' health promoting effects. Tilianin is a flavonoid glycoside that can be found in a wide range of medicinal plants (Khattulanuar et al., 2022) with reported benefits of being antioxidant (Nam et al., 2006), anti-inflammatory (Jiang et al., 2019), cardioprotective (Galvez et al., 2015) and neuroprotective (Yang et al., 2021). Despite its reported extensive biological activities and medicinal benefits, the bioactive form of tilianin and its mechanisms of action remain to be elucidated. Tilianin has a flavone backbone with a methyl group attached on its B-ring and a glucose group attached on its A-ring. Because of the functional groups located on its backbone and the double bond on its C-ring, tilianin is prone to *O*-deglycosylation, demethylation and hydrogenation. *O*-deglycosylation followed by demethylation yields apigenin (flavone) by way of acacetin. Hydrogenation of C2-C3 bonds of flavones leads to flavanone formation, which is also the first step for their metabolism through C-ring cleavage. Yang et al. identified a flavone reductase from *F. plautii* that catalyzes C-ring hydrogenation of apigenin to form naringenin, a common flavanone (Lagkouvardos et al., 2016). The diversity of metabolic reactions possible for tilianin make it an attractive candidate to study catabolism of dietary flavonoids by gut bacteria. Additionally, tilianin and its metabolites (acacetin, apigenin and naringenin) are all commercially available. Taken together, given its biological activity, chemical structure and commercial availability of metabolic derivatives, tilianin is an attractive model compound to investigate gut microbial metabolism of flavonoid glycosides and investigate whether flavonoid-derived metabolites elicit different physiological activities than their parent compounds.

For many dietary phytochemicals, the metabolic pathways in animals and their gut microbiota are not well characterized. One way to investigate these pathways is to reconstruct heterologous biosynthesis pathways using computational retro-biosynthesis tools (Jeffryes et al., 2018). These tools can mainly be classified into two approaches: knowledge-based approaches (Kuwahara et al., 2016) (Latendresse et al., 2014) that utilize reaction databases such as MetaCyc (Caspi et al., 2020), and rule-based approaches (Hafner et al., 2021) (Finnigan et al., 2021) that match the query molecule to generalized rules such as RetroPath RL (Koch et al., 2020). Knowledge-based tools have limited utility for more complex phytochemicals because reactions responsible for their metabolism are not comprehensively cataloged in databases. Rule-based methods utilize structural knowledge (e.g., subgraph patterns) of the compounds of interest to introduce a degree of generality to the reaction rules and thus are capable of predicting potentially novel metabolic pathways; however, this may lead to invalid predictions of products with structures unlikely found in nature (e.g., epoxides). More recently, tools that use machine learning methods to predict enzymatic reactions have shown the potential for higher performance and more accurate generalizations of reaction rules (Liu et al., 2017).

We evaluated our predictions against a reinforcement learning-based computational tool for predicting natural product biosynthetic pathways, BioNavi-NP. (Zheng et al., 2022 [↗](#)) BioNavi-NP uses a tree-based searching algorithm to generate retro-biosynthetic pathways and sorts the predicted reactions based on the computational prediction cost and pathway length. The computational cost reflects the confidence score, or a likelihood of a predicted reaction, where smaller cost means higher reaction probability. The reactions can be further ranked by the taxonomic distance between an optional user-defined organism and the organisms associated with the tool-predicted enzyme. Using default parameters, BioNavi-NP predicted that the most plausible transformation for tilianin is its O-deglycosylation to acacetin, demethylation of acacetin to apigenin, which overlaps with our RClass similarity-based predictions. The outputs from BioNavi-NP also suggested that acacetin can be methylated on its B-ring to apigenin 7,4-dimethyl ether, which can be demethylated on its A-ring to genkwanin. However, these reactions were flagged as likely implausible due to their lower confidence score. The most likely reaction pathway predicted by BioNavi-NP for apigenin is hydrogenation to naringenin followed by ring cleavage of naringenin to naringenin chalcone. The predicted compounds that are marked as “less likely” by BioNavi-NP were not detected in our in vitro metabolomics experiments.

One of the attractive features of BioNavi-NP is that it identifies potential enzymes that can catalyze the predicted reactions. It uses Selenzyme (Carbonell et al., 2018 [↗](#)) to rank the pathway output by reaction similarity and taxonomic distance between the organism of candidate enzyme and a user-defined organism. Like our RClass similarity-based analysis, BioNavi-NP also predicted that chalcone synthase can catalyze naringenin conversion to naringenin chalcone. However, it was unable to match the other predicted reactions with specific bacteria or bacterial enzymes. All other matched enzymes from BioNavi-NP’s output belong to known flavonoid pathways in plants. Our RClass similarity-based algorithm not only matches these transformations to known plant enzymes, it can also generate predictions for potential matches with bacterial enzymes.

Another useful feature of our algorithm is that its predictions are linked to functional similarity rather than sequence similarity. As shown in **Figure 2** [↗](#), an enzyme’s sequence similarity to another enzyme is not strongly correlated to their reaction class similarity. In this regard, relying on homology searches may limit discovery of enzymes that can catalyze reactions involving non-natural substrates through promiscuous enzyme activity.

Despite the above advantages, the prediction tool developed in this work has several limitations. The algorithm relies on the KEGG database to include the strains and enzymes of interest. For example, *Eubacterium ramulus*, a fully sequenced, known flavonoid-metabolizer strain, does not have a KEGG-identifier. Instead, it is recorded with its taxonomy ID (39490). Only three of its flavonoid relevant enzymes are cataloged: chalcone isomerase (CHI), flavonone/flavonol cleaving reductase (FCR) and phloretin hydrolase (PHY). Hence, a user cannot use this microorganism as an input to test its capability of metabolizing flavonoids through promiscuous enzyme activity. One solution to this problem is to use alternative databases such as the UniProt protein knowledgebase (UniProt, 2021 [↗](#)), to manually tabulate the enzymes that can be expressed by the organism of interest, and input this list instead of using the KEGG organism-identifier. We used this alternative approach to predict the reactions that can be catalyzed by *B. coccoides* as it also does not have a KEGG organism-identifier. Further, the flavanone/flavonol cleaving reductase from *E. ramulus* does not have an EC number despite its proven function of hydrogenating the C2-C3 bonds of flavanones and flavonols (Braune et al., 2019 [↗](#)). Yang et al. recently discovered a functionally similar enzyme, flavone reductase (FLR), in *F. plautii* (Yang et al., 2021 [↗](#)), which can initiate the gut microbial metabolism of flavones and flavonols by cleaving the C2-C3 bonds on the C-ring. Although *F. plautii* has a KEGG-identifier (fpla), the KEGG database does not report any homologous ene-reductases that are expressed by *F. plautii*. Hence, our tool failed to predict hydrogenation activity of *F. plautii* on flavones when “fpla” was used as input despite the organism’s proven ability to hydrogenate apigenin to naringenin. As FLR also does not have an EC

number, we were not able to use this enzyme as a single input. Instead, we had to input enoate reductase (EC: 1.3.1.31), a functionally similar ene-reductase, to predict the final step of the tilianin metabolism pathway.

In the present work, we demonstrated that the proposed tilianin pathway can be catalyzed by the select organisms: *B. animalis*, *B. coccoides* and *F. plautii*. In a comprehensive study by Lagkouvardos et al. (Lagkouvardos et al., 2016 [DOI](#)) all the 16s rRNA gene amplicon datasets from stool samples stored in Sequence Read Archive (SRA) were analyzed using the tool Integrative Microbial Next Generation Sequencing (IMNGS) (Leinonen et al., 2011 [DOI](#)). Their findings indicate that *B. animalis*, *B. coccoides* and *F. plautii* are shared by both human and mouse microbiomes. While *B. animalis* and *B. coccoides* are listed among the dominant strains in both organisms, the abundance of *F. plautii* is not very high. (Lagkouvardos et al., 2016 [DOI](#))

O-deglycosylation of tilianin to acacetin by *B. animalis* and demethylation of acacetin to apigenin by *B. coccoides* are novel findings facilitated by the use of the prediction tool developed in this work. Even though these two microbes can catalyze the predicted reactions individually, we cannot conclude that they together can catalyze the series of reactions solely based on the monoculture experiment results. For example, *B. animalis* can metabolize tilianin to acacetin, but transport of the metabolite acacetin to the extracellular space is not guaranteed. Further, it is unknown whether the amount of acacetin produced by *B. animalis* is enough for *B. coccoides* to utilize it to produce apigenin. In order to address these questions, we incubated the two microbes together and treated them with tilianin. After 48 hours of incubation, we observed a decrease in tilianin concentration and formation of acacetin and apigenin in the culture, indicating that *B. animalis* and *B. coccoides* together can metabolize tilianin all the way to apigenin.

Redox imbalance causing oxidative stress accumulation has been implicated as an important modulator of disorders involving cognitive decline. Extensive research exists around the use of H₂O₂ in in vitro neuronal cell injury models as a precursor to ROS, which invokes cell death through apoptotic processes (Wang et al., 2016 [DOI](#)). Here, we confirmed that PC-12 treatment with H₂O₂ induces cell death, likely through ROS mediated apoptosis. PC-12 cells provide a useful in vitro platform for understanding neurological response because of the extensive characterization of their catecholaminergic activity (Wiatrak et al., 2020 [DOI](#)). Emerging studies prove the utility of dietary polyphenols, namely flavonoids, as effective scavengers of ROS (Zhang & Tsao, 2016 [DOI](#)). However, radical scavenging potential may be associated with polyphenol structure and electron donation availability. For example, phenolic acids possessing acetic acid or propenoic acid groups achieved greater antioxidant activity than methoxy or hydroxy groups in the direct scavenging of 2, 2'-diphenyl-1-picrylhydrazyl (DPPH) and reduction of Fe(III) to Fe(II) (Chen et al., 2020 [DOI](#)). Similarly, flavonoid glycosides and aglycones differentially interact with cellular receptors such as the aryl hydrocarbon receptor (AhR), an important mediator of both prooxidant and antioxidant responses (Vrba et al., 2012 [DOI](#)). Thus, flavonoid protective activity could also rely on metabolic transformation. The present study determined differential protection against cell death attributed to ROS scavenging capacity of the parent glycoside and its metabolites. Interestingly, we noticed an inverse relationship between treatment dose and protective ability, suggesting potential toxicity and prooxidant behavior in high doses. This trend is consistent with a similar study investigating the protective capacity of luteolin against H₂O₂ induced injury of PC-12 cells, which determined concentrations of luteolin greater than 50 µM notably lowered cell viability (McGuigan et al., 1986 [DOI](#)). In addition to consideration of potential cellular toxicity, the concentrations of tilianin and metabolic pathway metabolites chosen for this study should also consider the concentration range present following human consumption. A study investigating excretion rates of green tea phenolic metabolites measured concentrations ranging from 1-7 µM of epicatechin and epigallocatechin, the primary flavonoids found in green tea, in human urine samples after administration of 1.2 g of green tea solids (Li et al., 2000 [DOI](#)). Thus, the selection of flavonoid concentrations in the presented study (i.e. 5 µM and 10µM) falls within a realistic range for human metabolism without imposing significant toxicity on the PC-12 test system. Further experiments

are warranted to fully characterize the concentration dependent ROS protective effects of phenolic metabolites. Nevertheless, the findings in this work clearly demonstrate a difference in biological activity elicited by a health promoting glycoside and its metabolites. We observed that metabolites that reduced loss of cell viability also reduced the level of ROS, suggesting cell viability is recovered through mitigation of oxidative stress. In contrast, lower viability is associated with increased formation of ROS, implying these metabolites are less effective scavengers.

In summary, this study presents a reaction class similarity-based prediction tool to identify the key taxa required to metabolize dietary flavonoids through investigating promiscuous enzyme activity. We report novel experimental evidence that *B. animalis* can *O*-deglycosylate tilianin to acacetin and *B. coccoides* can demethylate acacetin to apigenin, as predicted by the tool developed in this work. Combining our results with the earlier literature findings on apigenin hydrogenation to naringenin by *F. plautii*, we complete the metabolic pathway of tilianin by the gut microorganisms. We further showed that a flavonoid and its metabolites have varying potential to protect against cell death attributed to ROS scavenging capacity as antioxidants using an in vitro rat neuronal cell model. Taken together, these results will be beneficial in understanding the microbial metabolism-dependent biological effects of bioactive dietary flavonoids.

Experimental Procedures

Construction of the Prediction Algorithm

This project aims to develop a prediction-validation methodology to identify the key microorganisms that are responsible for flavonoid metabolism in the gut and predict potential promiscuous enzymes that can catalyze these reactions. Our prediction tool can take either a single enzyme in the form of Enzyme Commission (EC) number (e.g. “ec:2.1.1.75”), or a KEGG organism-identifier (e.g. “cpv”) or a consortium, a list of different organism-identifiers, as input. The output file consists of a table with possible reactions the input enzyme(s) or microorganism(s) can catalyze, the flavonoid substrates of these predicted reactions, and reaction likelihood score which is based on RClass similarity calculations. (See the **Supplementary Figure S1** [for an example input and output](#))

KEGG (Kyoto Encyclopedia of Genes and Genomes) is a knowledge-based integrated database for systematic analysis of genomic information with higher order functional information. It is classified into four major subcategories, which are Systems Information, Genomic Information, Chemical Information and Health Information. We utilized the Chemical Information subcategory to tabulate the relevant compound, reaction, reaction class (RClass) and enzyme data for our purposes. The tool was constructed in three steps. The first step is to tabulate all flavonoids and their first derivatives from the KEGG database. The second step is to retrieve their associated RClasses to further categorize them into major microbial flavonoid reactions. The last step is to create a master table that lists the calculated RClass similarity scores using the method developed by Muto et al. ([Muto et al., 2013](#) [for an example input and output](#))

Step 1: Tabulating flavonoids from the KEGG database

KEGG PATHWAY is a collection of manually drawn pathway maps representing the knowledge on molecular interactions. We searched map00941 (Flavonoid biosynthesis), map00942 (Anthocyanin biosynthesis), map00943 (Isoflavonoid biosynthesis) and map00944 (Flavone and flavonol biosynthesis) pathways to collect the flavonoids and their first neighbors, as these neighbors represent the closest derivatives of chosen flavonoids. Next, we searched the Phytochemical Compounds list from the KEGG database to retrieve the compounds that are not available in the KEGG PATHWAY collection. We removed aurones, 3-arylcoumarins, pterocarpanes, coumestans, malonates, rotenones, 2-aryl benzofurans and complex flavonoids as they do not carry the key

flavonoid structure which consists of two phenyl rings and a heterocyclic C-ring containing the embedded oxygen. In the end, we collected 312 unique flavonoids and flavonoid metabolites. (**Supplementary Table S1** [↗](#))

Step 2: Retrieving and classifying the associated RClasses

KEGG RClass contains reaction classifications based on the chemical transformation patterns that are represented as RDM patterns. The RDM patterns are defined based on the reaction center (R), difference region (D) and matched region (M) for each substrate-product pair. RClasses represent each unique RDM pattern or a combination of RDM patterns if more than one reaction center is identified for a chemical transformation. Each RClass is identified by the RC number and represented by KEGG atom types. For example, R09791, methylation reaction of naringenin by naringenin 7-O-methyltransferase, belongs to the RClass RC00392. Its KEGG atom type illustration is O1a-O2a:*-C1a:C8y-C8y, which represents the RDM pattern of this particular reaction. The KEGG atom type usually consists of three characters: the first character, the upper case letter, represents the atomic species (O: oxygen), the second character, the number, represents the primary bonding, and the third character, the lower case letter, indicates the topological information such as the number of substituted group. “*” is used in place of hydrogen atoms.

In this step, we aim to retrieve and classify the RClasses that represent microbial flavonoid reactions that can modify our tabulated 312 unique flavonoids and their derivatives. KEGG API is a REST-style Application Programming Interface to the KEGG database resource. It allows KEGG users to retrieve information for databases of their interest and link related entries by using database cross-references. There is not a direct way to retrieve RClasses from the compound IDs using the KEGG API feature. Thus, one needs to obtain the enzymes that can utilize a compound of interest as substrates first, then link them to their corresponding RClasses. Using this two-step procedure, we first collected the enzymes that are associated with the selected flavonoids and their metabolites in the form of EC numbers. We obtained 100 unique enzymes and added an optional feature to eliminate the ones that catalyze reactions with molecular oxygen, as the majority of the gut microbiome is comprised of obligate anaerobic bacteria (**Supplementary Table S2** [↗](#) & **S3** [↗](#)). Using the tabulated enzymes, we retrieved the corresponding RClasses. We identified 19 unique RClasses that represent the major flavonoid reactions, which are O-deglycosylation, C-hydrolysis, de/hydrogenation, di/hydroxylation, ring cleavage, anthocyanin/anthocyanidin di/hydroxylation and anthocyanin/anthocyanidin de/hydrogenation reactions. We separated the di/hydroxylation and de/hydrogenation reactions for the C-rings of anthocyanins and anthocyanidins, because their reaction centers differ from other flavonoids due to their positive charge. All reactions were considered reversible unless specified otherwise.

Step 3: Listing the calculated RClass similarity scores

The algorithm predicts promiscuous reactions by comparing the RClasses. As KEGG RClasses are represented using the KEGG atom type format, instead of more conventional identifiers such as InChIKey or SMILES, available chemical similarity clustering cheminformatics toolkits are unable to calculate the similarity scores between different RClasses. Thus, we utilized the RClass chemical similarity calculation method developed by Muto et al. ([Muto et al., 2013](#) [↗](#)). Briefly, their method divides RDM representation into binary vectors where each atom type is mapped based on their fingerprints. Fingerprints are represented using letters and they indicate presence or absence of a particular atom, functional groups or bond type. Then those binary vectors are used to calculate the Jaccard similarity coefficient between different RClasses. Calculated scores are normalized to range between 0 and 1. The KEGG database uses this method internally to create a database for pairwise RClass similarity comparisons for each RClass that is listed. We exported those pairwise similarity scores for our 19 unique RClasses that represent the major flavonoid reactions to create a master table (**Supplementary Table S4** [↗](#)). As a result, once the algorithm takes an enzyme, or a list of enzymes derived from an organism ID (or multiple organism IDs), it retrieves the RClasses that are associated with the input enzymes. Then it finds whether these RClasses are similar to the

key RClasses that represent microbial flavonoid metabolism reactions by searching them in the master RClass similarity table. If the similarity score is above the cutoff value of 0.8, the algorithm then retrieves the associated compounds using the KEGG API feature and refine them to only get flavonoids. We chose our cutoff value as 0.8, because when we applied the algorithm to the known flavonoid-metabolizing enzymes, we found that their similarity scores fall above this value as shown in the Results section. Finally, the sorted output contains the reaction type, similarity score and potential flavonoid substrates.

Prediction of Promiscuous Enzyme Activity of Flavonoid Metabolism

As a proof of concept, we aimed to predict the gut microbial flavonoid reactions and validate these predictions using in vitro culturing methods and targeted metabolomics experiments. We chose tilianin, a flavone glycoside, as the parent compound because it carries a large variety of functional groups and it is commercially available in its pure form along with its proposed metabolites. We proposed a pathway of O-deglycosylation of tilianin to acacetin, aglycone version of tilianin, demethylation of acacetin to apigenin, hydrogenation of apigenin to naringenin and ring cleavage reaction of naringenin to 3-(4-hydroxyphenyl) propionic acid. Naringenin ring cleavage was heavily investigated in a recent study by Gülşan et al. (Gülşan et al., 2022 [DOI](#)), hence, we excluded this reaction from our analysis. We started our predictions with educated guesses. Bifidobacteriaceae species have been repeatedly reported to remove the sugar moiety from flavonoid glycosides under anaerobic conditions (Marotti et al., 2007 [DOI](#))). *B. animalis* is a well-studied, fully sequenced, commercially available gut anaerobe, hence, we decided to investigate the predictions on this strain. Our algorithm predicted 38 flavonoid metabolizing reactions for *B. animalis*, 30 of them being O-deglycosylation reactions. Recent studies report that *B. animalis* can catalyze the O-deglycosylation of anthocyanidins and isoflavones (Ávila et al., 2009 [DOI](#)), but its activity on flavones remains to be elucidated. Our algorithm predicts that this strain can perform O-deglycosylation reaction on vitexin, prunin, hesperetin 7-O-Glucoside, neo-hesperidin and apigetrin. Our predictions are limited to the compounds that are tabulated by the KEGG database. Because tilianin does not have a KEGG identifier, it is expected that our algorithm cannot predict *B. animalis* metabolic activity on tilianin. Apigetrin, on the other hand, is a demethylation product of tilianin whose sugar moiety, A and C rings are identical to tilianin. *B. animalis* O-deglycosylation activity on apigetrin was predicted by our algorithm, which gives us confidence that this strain can also metabolize tilianin to its aglycone form, acacetin.

We followed a similar methodology to make predictions for the demethylation reaction. *Blautia* sp. MRG-PMF1 strain was shown to catalyze demethylation of hesperetin (a flavanone) to eriodictyol and biochanin A (an isoflavone) to genistein (Kim et al., 2014 [DOI](#)). Similar to *B. animalis*, *B. coccoides*, from the same genera with *Blautia* sp. MRG-PMF1, is a fully sequenced common gut anaerobe. Thus, we ran our algorithm using this strain. The algorithm predicts that it can demethylate acacetin to apigenin, thus, we decided to test *B. coccoides* strain for this reaction.

Tabulating and Comparing Sequence and Functional Similarity Scores of Predicted Enzymes

The sequence similarity of a variety of enzymes from *E. coli*, *F. plautii*, mouse, mouse ear cress, and human are plotted against their functional similarity to demonstrate that sequence similarity does not correlate to functional similarity. The sequence similarities of enzymes were obtained through BLAST® from National Library of Medicine (NIH), and the percent identity in blast result is used as the sequence similarity for the enzymes. DB Search feature of the KEGG database were used to tabulate the functional similarity scores of RClasses that are enabled by the selected enzymes. These sequence similarity scores of each enzyme were then tabulated alongside with the functional similarity scores between all other enzymes of the same reaction class and organism, excluding comparisons with themselves. These data points were then graphically represented by

bubble plots. Bubble plots were chosen because there are a significant portion of enzymes that have identical sequence and functional similarity scores. For example, there are 2926 unique enzyme pairing in human *O*-deglycosylation group, but there are only 58 unique sequence and functional similarity score pairing. This calls for another dimension in addition to the x and y axis to showcase the degree of repeats, which is the size of the bubbles (data points). All data points, unique or otherwise, were then used to calculate the r and p values. This was done in Python with the `pearsonr` function in the SciPy package.

Cell Lines, Bacterial Strains, Culture Conditions, and Reagents

B. animalis ATCC 700541, *B. coccoides* ATCC 29236, *F. plautii* ATCC 49531 were obtained from ATCC (Manassas, VA). *B. animalis* and *B. coccoides* were cultured in brain heart infusion (BHI) broth (Becton, Dickinson and Company, Franklin Lakes, NJ) supplemented with 0.5% yeast extract (Becton, Dickinson and Company, Franklin Lakes, NJ), 0.05% cysteine (Alfa Aesar, Tewksbury, MA), 0.05% hemin, and 0.02% vitamin K (Sigma-Aldrich, St. Louis MO). *F. plautii* was grown in Wilkens-Chalgren Anaerobe (WCA) Broth (Thermo Fisher Scientific, Waltham, MA). Tilianin, acacetin, apigenin, and naringenin (HPLC grade, ≥95.0% purity) were purchased from Sigma-Aldrich (St. Louis, MO). All other high-purity LCMS-grade reagents were also purchased from Sigma Aldrich unless specified otherwise.

Bacterial Monoculture and Co-culture

Bacterial monoculture experiments with strict anaerobes *B. animalis*, *B. coccoides*, *F. plautii* and *E. coli* were carried out in an anaerobic chamber with 85% N₂, 10% CO₂, 5% H₂ (Coy Lab, Grass Lake, MI) in their respective media mentioned above. All reagents and media were introduced into the anaerobic chamber a day before the experiment to ensure that they were pre-reduced by the start of the experiment. Cultures grown to an OD (absorbance at 600 nm) of 1.0 were diluted 10-fold for *B. animalis* and *B. coccoides* and treated with 100 μM of tilianin and acacetin, respectively (stock solutions were prepared using DMSO as solvent and then filter sterilized). *F. plautii* monoculture was treated with 100 μM apigenin without any dilution due to its faint growth. Monocultures treated with 0.1% v/v DMSO were used as vehicle controls. Following the flavonoid or vehicle treatment, monocultures were incubated at 37 °C for 48 h. Each monoculture experiment was run in triplicate. Samples were collected for metabolite extraction by sacrificing the cultures after 24 h of incubation.

For bacterial co-culture, *B. animalis* and *B. coccoides* cultures grown to an OD₆₀₀ of 1.0 were pelleted down by centrifugation at 10,000g for 15 min (Eppendorf 5424R) and resuspended in 2.5mL of BHI each and then put together (final culture volume is 5 mL). This co-culture sample was then treated with 100 μM of tilianin and incubated at 37 °C for next 48 h. Each co-culture experiment was run in triplicates under anaerobiosis. Samples were collected for metabolite extraction by sacrificing the cultures after 24 and 48 h of incubation.

For the co-culture of *B. animalis* and *F. plautii*, monocultures grown to an OD₆₀₀ of 0.2 were pelleted down by centrifugation at 10,000g for 15 min (Eppendorf 5424R) and resuspended in 2.5mL of BHI each and then put together (final culture volume is 5 mL). The OD was comparatively lower due to the faint growth of *F. plautii* even after 24h of culture. This co-culture samples were then treated with 100 μM of tilianin and incubated at 37 °C for next 48 h. Each co-culture experiment was run in triplicates under anaerobiosis. Co-cultures treated with 0.1% v/v DMSO and maintained in the same conditions were used as vehicle controls. Both treatment and control samples were collected for metabolite extraction by sacrificing the cultures after 24 and 48 h of incubation.

Metabolite Extraction

Metabolites were extracted from the harvested cell culture samples using a solvent-based method. Briefly, 50 μL of culture sample was mixed with 200 μL of ice-cold methanol using a vortex mixer and kept on ice for 60 minutes. The sample-solvent mixture was then centrifuged at 15,000g at 4 °C for 5 min. The supernatant was filtered through a 0.22 μm sterile nylon filter (Corning). The filtrate was then passed through a 3 kDa molecular weight cutoff filter (Amicon Ultra, Sigma-Aldrich) placed in a collection tube by centrifuging the sample tube at 15,000g for 45 min at 4 °C. The filtrate from the collection tube was then mixed with 2 mL sterile water, lyophilized to a pellet using a freeze dryer (FreeZone, Labconco Corporation) and stored at -80 °C until further analysis. Prior to LC-MS/MS analysis, the freeze-dried samples were reconstituted in 250 μL of 50% (v/v) methanol/water.

Targeted Analysis of Tilianin Metabolites

The extracted samples were analyzed for tilianin, acacetin, apigenin and naringenin using product ion scan experiments performed on a triple-quadrupole time-of-flight instrument (5600+, AB Sciex) coupled to a binary pump high-performance liquid chromatography (HPLC) system (1260 Infinity, Agilent). Tilianin was detected in positive ionization mode, while its metabolites were detected using negative ionization mode. Chromatographic separation was achieved on a Synergi 4 μm Hydro, 250 mm $\times$ 2 mm, 80 Å reverse phase column (Phenomenex, Torrance, CA) maintained at 30 °C using a solvent gradient method. Solvent A was 0.2% (v/v) formic acid in water and solvent B was 100% acetonitrile. The gradient method was as follows: 0–30 min (95% A to 45% A), 30–38 min (45% A to 95% A). The flow rate was set to 0.6 mL/min. The injection volume was 10 μL . Raw data were processed in PeakView 2.1 to determine the ion peaks. The peak areas in the extracted ion chromatograms were manually integrated using MultiQuant 2.1 (AB Sciex). The identities of detected compounds were confirmed by matching their retention time and/or MS/MS spectra to high-purity standards run on the same instrument using the same method. De/methylated and di/hydroxylated apigenin and naringenin metabolites and the compounds predicted by BioNavi-NP were only searched by their monoisotopic masses in time-of-flight experiments and were not confirmed with their high-purity standards.

Investigation of in vitro Protective Effect of Flavonoids in a Neuronal Cell Model

Cell Culture

Rat adrenal pheochromocytoma PC-12 cell lines were obtained from ATCC (CRL-1721). Cells were cultured at 37 °C at 5% CO_2 . Cells were cultured in Rosewell Park Memorial Institute 1640 medium (RPMI, ThermoFisher Scientific, Waltham, MA) supplemented with 10% horse serum (ThermoFisher Scientific, Waltham, MA), 5% FBS (ATCC, Manassas, VA), 50 U/mL Penicillin-streptomycin (ThermoFisher Scientific, Waltham, MA), and 2 g/L of NaHCO_3 (Sigma-Aldrich, St. Louis, MO). Two thirds of the medium was exchanged every other day for fresh medium and cells were passaged approximately every 4 days. All well plates and culture flasks were coated prior to seeding with 7 $\mu\text{g/mL}$ rat tail tendon collagen I (Sigma-Aldrich, St. Louis, MO) and rinsed with PBS 3 times. PC-12 cells used for experiments were in the exponential growth phase of cell growth between passage 4-6.

PC-12 cells were seeded into 96-well plates coated with collagen I (ThermoFisher, Waltham, MA) and rinsed with 3 volumes of PBS. The cells were seeded at a density of 5×10^5 cells/mL then allowed to adhere for 24 hours. PC-12 cells were then cultured in fresh medium containing 0 μM , 100 μM , 200 μM , 500 μM H_2O_2 (Sigma-Aldrich, St. Louis, MO) for 4 hours to determine the optimal stress concentration for subsequent experiments (data not shown). Based on cell survival rate, 200 μM H_2O_2 was chosen as the oxidative stress concentration.

PC-12 Viability Assay

PC-12 cells were seeded in collagen I coated 96-well plates at 5×10^5 cells/mL and allowed to adhere for 24 hours. Cells were then pretreated with fresh medium containing 5 μ M or 10 μ M of either tilianin or tilianin pathway metabolite in quadruplicate. The control conditions received only fresh medium without pretreatment metabolite or 1mM N-acetylcysteine (NAC, Sigma-Aldrich, St. Louis, MO) as the positive control. Cells were pretreated for 1 hour then exposed to 200 μ M H_2O_2 for 4 hours. Cell viability was determined using the CellTrace™ Calcein violet live cell stain (ThermoFisher, Waltham, MA). The cell membrane permeable non-fluorescent calcein AM is cleaved by nonspecific intracellular esterase to form highly fluorescent calcein detected at an excitation wavelength and emission wavelength of 400 nm and 452 nm respectively. Cells were first washed twice with PBS then stained with 5 μ M calcein in the dark for 1 hour. Fluorescence measurements were then taken using a SpectraMax M3 Multi-Mode Microplate Reader (Molecular Devices, San Jose, CA).

Detection of Reactive Oxygen Species

The 2,7-dichlorodihydrofluorescein diacetate (H_2DCFDA , Sigma-Aldrich, St. Louis, MO) probe was used to measure reactive oxygen species. The cell permeable nonfluorescent H_2DCFDA is cleaved by intracellular esterase and oxidized to highly fluorescent dichlorofluorescein (DCF) detected with an excitation wavelength and emission wavelength of 492 nm and 517 nm respectively. The measurement of reactive oxygen species followed the same procedure used to determine cell viability. Briefly, PC-12 cells were seeded into 96-well plates and cultured for 24 hours for adherence to the collagen I coating. Cells were then pretreated with 5 μ M or 10 μ M of either tilianin or tilianin pathway metabolite for 1 hour in quadruplicate. Experimental controls received only fresh medium or pretreatment with 1mM NAC for 1 hour. The cells were then stressed with 200 μ M H_2O_2 for 4 hours. Treated cells were washed twice with PBS and stained with 20 μ M H_2DCFDA in the dark for 1 hour. Fluorescence was measured using the SpectraMax M3 Multi-Mode Microplate Reader (Molecular Devices, San Jose, CA).

Acknowledgements

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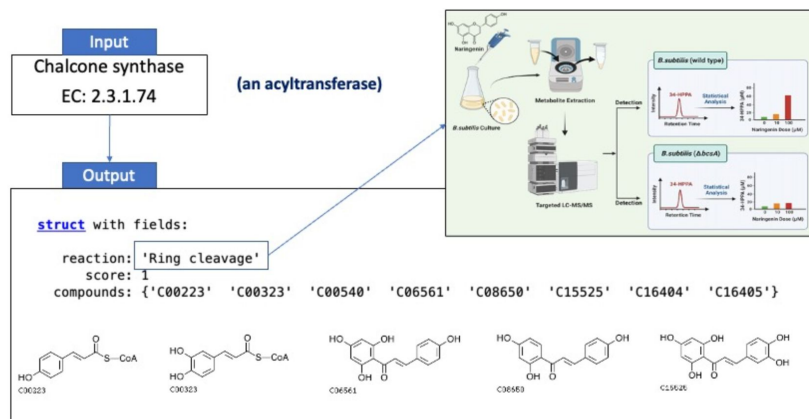


Figure S1

Example input and output from the tool developed in this work

We used chalcone synthase enzyme with the EC number 2.3.1.74 as an input. The output suggests that this input enzyme can perform ring cleavage reaction on flavonoids and their metabolites, which aligns with our findings from Chapter 2.

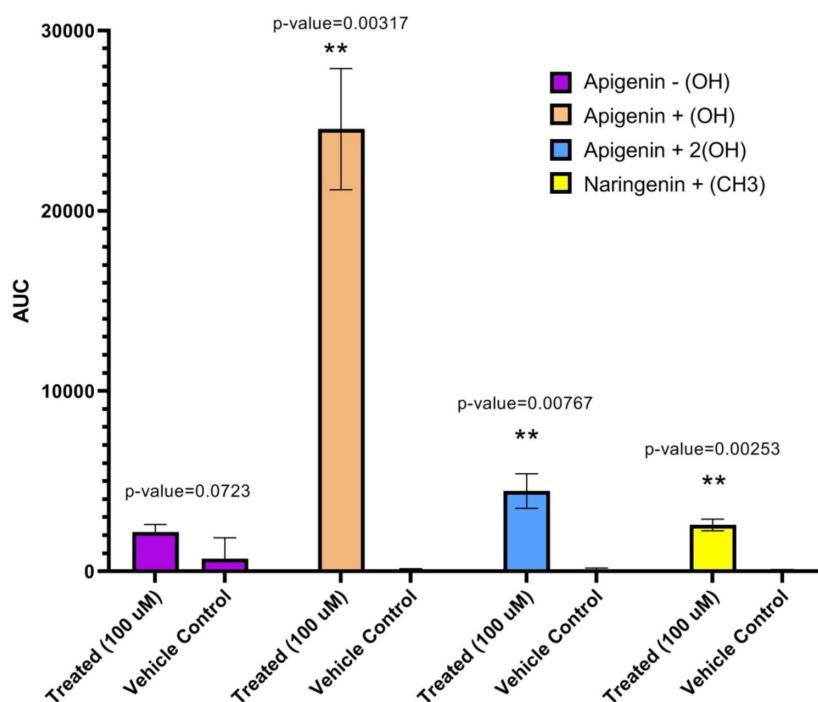


Figure S2

Concentrations of possible microbial metabolites of apigenin and naringenin detected in the apigenin treated *B. animalis* monoculture.

Area under the curve for the detected apigenin and naringenin metabolites. Double sterisk (**) indicates significant difference ($p < 0.005$) compared to the vehicle control (no apigenin).

C00223	C05903	C09755	C10097	C10507	C14313
C00323	C05904	C09756	C10098	C10509	C14314
C00389	C05905	C09757	C10102	C10510	C14344
C00509	C05906	C09762	C10104	C10513	C14472
C00540	C05907	C09770	C10105	C10520	C14536
C00774	C05908	C09774	C10107	C10521	C14942
C00786	C05909	C09789	C10108	C10524	C14953
C00814	C05911	C09793	C10109	C10526	C14985
C00852	C05990	C09803	C10110	C10527	C15019
C00858	C06561	C09805	C10111	C10530	C15109
C00974	C06562	C09806	C10112	C10533	C15118
C01265	C06563	C09807	C10113	C10534	C15509
C01378	C08064	C09826	C10114	C10536	C15525
C01460	C08574	C09827	C10115	C10538	C15531
C01470	C08578	C09828	C10116	C10540	C15567
C01477	C08600	C09830	C10117	C11264	C15608
C01514	C08604	C09833	C10118	C11577	C16187
C01562	C08620	C09939	C10119	C11620	C16188
C01604	C08629	C09982	C10120	C11621	C16190
C01617	C08639	C10019	C10121	C12123	C16195
C01709	C08647	C10021	C10122	C12124	C16225
C01714	C08649	C10022	C10175	C12125	C16232
C01750	C08650	C10023	C10176	C12127	C16233
C01821	C08652	C10024	C10177	C12128	C16298
C02495	C08716	C10025	C10178	C12134	C16305
C02549	C08718	C10026	C10180	C12135	C16306
C02675	C08724	C10028	C10181	C12136	C16307
C02806	C08725	C10029	C10183	C12137	C16312
C02906	C08729	C10030	C10184	C12138	C16314
C02920	C09099	C10031	C10185	C12139	C16315
C03515	C09126	C10032	C10186	C12140	C16404
C03567	C09318	C10033	C10187	C12208	C16405
C03648	C09320	C10036	C10188	C12249	C16406
C03870	C09471	C10037	C10189	C12625	C16407
C03951	C09478	C10038	C10190	C12626	C16408

Table S1.

Tabulated flavonoids and their derivatives (KEGG COMPOUND numbers)

C04024	C09505	C10039	C10191	C12627	C16409
C04109	C09614	C10040	C10192	C12628	C16410
C04293	C09616	C10041	C10193	C12629	C16415
C04443	C09617	C10043	C10195	C12630	C16416
C04444	C09618	C10044	C10196	C12631	C16417
C04491	C09636	C10045	C10197	C12633	C16418
C04527	C09641	C10046	C10199	C12634	C16419
C04552	C09644	C10047	C10200	C12635	C16422
C04581	C09646	C10049	C10203	C12636	C16490
C04608	C09650	C10050	C10205	C12638	C16492
C05334	C09727	C10058	C10208	C12639	C16911
C05376	C09728	C10068	C10216	C12644	C19727
C05622	C09732	C10072	C10415	C12667	C19796
C05623	C09734	C10073	C10419	C12668	C21833
C05625	C09735	C10078	C10420	C14079	C21854
C05631	C09736	C10079	C10465	C14131	C21913
C05902	C09751	C10084	C10471	C14290	C22048

Table S1. (continued)

ec:1.1.1.219	ec:1.3.1.77	ec:2.3.1.116	ec:2.4.1.254
ec:1.1.1.234	ec:2.1.1.104	ec:2.3.1.133	ec:2.4.1.286
ec:1.1.1.348	ec:2.1.1.150	ec:2.3.1.153	ec:2.4.1.294
ec:1.14.11.9	ec:2.1.1.153	ec:2.3.1.170	ec:2.4.1.297
ec:1.14.14.135	ec:2.1.1.155	ec:2.3.1.171	ec:2.4.1.298
ec:1.14.14.81	ec:2.1.1.169	ec:2.3.1.173	ec:2.4.1.299
ec:1.14.14.82	ec:2.1.1.175	ec:2.3.1.215	ec:2.4.1.300
ec:1.14.14.87	ec:2.1.1.212	ec:2.3.1.74	ec:2.4.1.357
ec:1.14.14.88	ec:2.1.1.231	ec:2.4.1.105	ec:2.4.1.81
ec:1.14.14.89	ec:2.1.1.232	ec:2.4.1.106	ec:2.4.1.91
ec:1.14.14.90	ec:2.1.1.254	ec:2.4.1.115	ec:2.4.2.25
ec:1.14.14.91	ec:2.1.1.267	ec:2.4.1.116	ec:2.4.2.35
ec:1.14.14.93	ec:2.1.1.270	ec:2.4.1.159	ec:2.4.2.50
ec:1.14.14.96	ec:2.1.1.338	ec:2.4.1.170	ec:2.4.2.51
ec:1.14.19.63	ec:2.1.1.339	ec:2.4.1.185	ec:2.5.1.36
ec:1.14.19.76	ec:2.1.1.42	ec:2.4.1.189	ec:2.8.2.25
ec:1.14.20.4	ec:2.1.1.46	ec:2.4.1.190	ec:2.8.2.26
ec:1.14.20.5	ec:2.1.1.75	ec:2.4.1.191	ec:2.8.2.27
ec:1.14.20.6	ec:2.1.1.76	ec:2.4.1.234	ec:2.8.2.28
ec:1.17.1.3	ec:2.1.1.78	ec:2.4.1.236	ec:3.2.1.167
ec:1.21.3.6	ec:2.1.1.82	ec:2.4.1.237	ec:3.2.1.168
ec:1.3.1.117	ec:2.1.1.83	ec:2.4.1.238	ec:3.2.1.31
ec:1.3.1.45	ec:2.1.1.84	ec:2.4.1.239	ec:4.2.1.105
ec:1.3.1.46	ec:2.1.1.88	ec:2.4.1.240	ec:4.2.1.139
ec:1.3.1.51	ec:2.3.1.115	ec:2.4.1.253	ec:5.5.1.6

Table S2.

Tabulated enzymes from the chosen flavonoids and their derivatives

ec:1.1.1.170	ec:1.14.13.122	ec:1.14.99.57	ec:1.14.13.16	ec:1.14.20.4	ec:1.14.11.67	ec:1.14.19.21	ec:1.14.11.12	ec:1.14.15.35
ec:1.1.3.5	ec:1.14.13.131	ec:1.14.99.61	ec:1.14.13.22	ec:1.14.20.8	ec:1.14.11.71	ec:1.14.19.32	ec:1.14.11.17	ec:1.14.15.39
ec:1.1.3.9	ec:1.14.13.148	ec:1.14.99.65	ec:1.14.13.27	ec:1.14.20.12	ec:1.14.11.75	ec:1.14.19.44	ec:1.14.11.24	ec:1.14.16.4
ec:1.1.3.13	ec:1.14.13.160	ec:1.14.99.-	ec:1.14.13.33	ec:1.14.20.-	ec:1.14.12.1	ec:1.14.19.51	ec:1.14.11.28	ec:1.14.16.-
ec:1.1.3.17	ec:1.14.13.168	ec:1.16.3.3	ec:1.14.13.39	ec:1.14.99.4	ec:1.14.12.9	ec:1.14.19.55	ec:1.14.11.36	ec:1.14.18.1
ec:1.1.3.21	ec:1.14.13.178	ec:1.17.3.4	ec:1.14.13.46	ec:1.14.99.15	ec:1.14.12.13	ec:1.14.19.59	ec:1.14.11.40	ec:1.14.18.5
ec:1.1.3.29	ec:1.14.13.182	ec:1.21.3.4	ec:1.14.13.58	ec:1.14.99.23	ec:1.14.12.17	ec:1.14.19.63	ec:1.14.11.44	ec:1.14.18.10
ec:1.1.3.39	ec:1.14.13.196	ec:1.21.3.8	ec:1.14.13.63	ec:1.14.99.34	ec:1.14.12.23	ec:1.14.19.67	ec:1.14.11.49	ec:1.14.19.1
ec:1.1.3.43	ec:1.14.13.210	ec:4.1.1.90	ec:1.14.13.81	ec:1.14.99.44	ec:1.14.12.-	ec:1.14.19.71	ec:1.14.11.56	ec:1.14.19.8
ec:1.1.3.47	ec:1.14.13.216	ec:4.1.1.-	ec:1.14.13.92	ec:1.14.99.50	ec:1.14.13.5	ec:1.14.19.75	ec:1.14.11.60	ec:1.14.19.12
ec:1.1.5.4	ec:1.14.13.220	ec:7.1.1.5	ec:1.14.13.111	ec:1.14.99.54	ec:1.14.13.9	ec:1.14.19.-	ec:1.14.11.64	ec:1.14.19.18
ec:1.2.3.1	ec:1.14.13.225	ec:1.1.1.270	ec:1.14.13.127	ec:1.14.99.58	ec:1.14.13.18	ec:1.14.20.5	ec:1.14.11.68	ec:1.14.19.25
ec:1.2.3.6	ec:1.14.13.229	ec:1.1.3.6	ec:1.14.13.135	ec:1.14.99.62	ec:1.14.13.23	ec:1.14.20.9	ec:1.14.11.72	ec:1.14.19.37
ec:1.2.3.13	ec:1.14.13.233	ec:1.1.3.10	ec:1.14.13.149	ec:1.14.99.67	ec:1.14.13.29	ec:1.14.20.13	ec:1.14.11.76	ec:1.14.19.48
ec:1.3.3.3	ec:1.14.13.237	ec:1.1.3.14	ec:1.14.13.161	ec:1.14.-.-	ec:1.14.13.34	ec:1.14.21.-	ec:1.14.12.3	ec:1.14.19.52
ec:1.3.3.7	ec:1.14.13.241	ec:1.1.3.18	ec:1.14.13.170	ec:1.17.3.1	ec:1.14.13.40	ec:1.14.99.11	ec:1.14.12.10	ec:1.14.19.56
ec:1.3.3.12	ec:1.14.13.245	ec:1.1.3.23	ec:1.14.13.179	ec:1.21.3.1	ec:1.14.13.50	ec:1.14.99.20	ec:1.14.12.14	ec:1.14.19.60
ec:1.3.3.16	ec:1.14.14.1	ec:1.1.3.30	ec:1.14.13.187	ec:1.21.3.5	ec:1.14.13.59	ec:1.14.99.24	ec:1.14.12.18	ec:1.14.19.64
ec:1.3.99.30	ec:1.14.14.9	ec:1.1.3.40	ec:1.14.13.200	ec:1.21.98.2	ec:1.14.13.64	ec:1.14.99.35	ec:1.14.12.24	ec:1.14.19.68
ec:1.4.3.2	ec:1.14.14.13	ec:1.1.3.44	ec:1.14.13.211	ec:4.1.1.107	ec:1.14.13.82	ec:1.14.99.46	ec:1.14.13.1	ec:1.14.19.72
ec:1.4.3.7	ec:1.14.14.17	ec:1.1.3.48	ec:1.14.13.217	ec:4.1.99.5	ec:1.14.13.101	ec:1.14.99.51	ec:1.14.13.6	ec:1.14.19.76
ec:1.4.3.12	ec:1.14.14.21	ec:1.1.99.9	ec:1.14.13.222	ec:7.1.1.7	ec:1.14.13.113	ec:1.14.99.55	ec:1.14.13.10	ec:1.14.20.1
ec:1.4.3.16	ec:1.14.14.25	ec:1.2.3.3	ec:1.14.13.226	ec:1.1.3.2	ec:1.14.13.128	ec:1.14.99.59	ec:1.14.13.19	ec:1.14.20.6
ec:1.4.3.22	ec:1.14.14.30	ec:1.2.3.7	ec:1.14.13.230	ec:1.1.3.7	ec:1.14.13.146	ec:1.14.99.63	ec:1.14.13.24	ec:1.14.20.10
ec:1.4.3.26	ec:1.14.14.34	ec:1.2.3.14	ec:1.14.13.234	ec:1.1.3.11	ec:1.14.13.153	ec:1.14.99.68	ec:1.14.13.31	ec:1.14.20.14
ec:1.5.3.2	ec:1.14.14.38	ec:1.3.3.4	ec:1.14.13.238	ec:1.1.3.15	ec:1.14.13.163	ec:1.15.1.1	ec:1.14.13.35	ec:1.14.99.1
ec:1.5.3.7	ec:1.14.14.42	ec:1.3.3.8	ec:1.14.13.242	ec:1.1.3.19	ec:1.14.13.171	ec:1.17.3.2	ec:1.14.13.43	ec:1.14.99.12
ec:1.5.3.14	ec:1.14.14.46	ec:1.3.3.13	ec:1.14.13.246	ec:1.1.3.27	ec:1.14.13.180	ec:1.21.3.2	ec:1.14.13.51	ec:1.14.99.21
ec:1.5.3.18	ec:1.14.14.50	ec:1.3.5.6	ec:1.14.14.3	ec:1.1.3.37	ec:1.14.13.189	ec:1.21.3.6	ec:1.14.13.61	ec:1.14.99.26
ec:1.5.3.23	ec:1.14.14.54	ec:1.3.99.31	ec:1.14.14.10	ec:1.1.3.41	ec:1.14.13.208	ec:2.6.1.114	ec:1.14.13.66	ec:1.14.99.38
ec:1.6.3.2	ec:1.14.14.58	ec:1.4.3.3	ec:1.14.14.14	ec:1.1.3.45	ec:1.14.13.212	ec:4.1.1.108	ec:1.14.13.83	ec:1.14.99.47
ec:1.7.2.1	ec:1.14.14.62	ec:1.4.3.8	ec:1.14.14.18	ec:1.1.3.49	ec:1.14.13.218	ec:7.1.1.3	ec:1.14.13.105	ec:1.14.99.52

Table S3.

Enzymes that catalyze reactions which require molecular oxygen

ec:1.7.3.5	ec:1.14.14.66	ec:1.4.3.13	ec:1.14.14.22	ec:1.1.99.18	ec:1.14.13.22 3	ec:7.1.1.9	ec:1.14.13.11 4	ec:1.14.99.5 6
ec:1.8.3.3	ec:1.14.14.70	ec:1.4.3.19	ec:1.14.14.26	ec:1.2.3.4	ec:1.14.13.22 7	ec:1.1.3.4	ec:1.14.13.13 0	ec:1.14.99.6 0
ec:1.8.3.7	ec:1.14.14.74	ec:1.4.3.23	ec:1.14.14.31	ec:1.2.3.8	ec:1.14.13.23 1	ec:1.1.3.8	ec:1.14.13.14 7	ec:1.14.99.6 4
ec:1.10.3.4	ec:1.14.14.78	ec:1.4.3.-	ec:1.14.14.35	ec:1.2.3.15	ec:1.14.13.23 5	ec:1.1.3.12	ec:1.14.13.15 4	ec:1.14.99.6 9
ec:1.10.3.11	ec:1.14.14.82	ec:1.5.3.4	ec:1.14.14.39	ec:1.3.3.5	ec:1.14.13.23 9	ec:1.1.3.16	ec:1.14.13.16 7	ec:1.16.3.1
ec:1.10.3.-	ec:1.14.14.86	ec:1.5.3.10	ec:1.14.14.43	ec:1.3.3.10	ec:1.14.13.24 3	ec:1.1.3.20	ec:1.14.13.17 2	ec:1.17.3.3
ec:1.13.11.2	ec:1.14.14.90	ec:1.5.3.15	ec:1.14.14.47	ec:1.3.3.14	ec:1.14.13.24 7	ec:1.1.3.28	ec:1.14.13.18 1	ec:1.21.3.3
ec:1.13.11.6	ec:1.14.14.94	ec:1.5.3.19	ec:1.14.14.51	ec:1.3.99.26	ec:1.14.14.5	ec:1.1.3.38	ec:1.14.13.19 5	ec:1.21.3.7
ec:1.13.11.11	ec:1.14.14.98	ec:1.5.3.-	ec:1.14.14.55	ec:1.3.-.-	ec:1.14.14.11	ec:1.1.3.42	ec:1.14.13.20 9	ec:4.1.1.39
ec:1.13.11.16	ec:1.14.14.102	ec:1.6.3.3	ec:1.14.14.59	ec:1.4.3.4	ec:1.14.14.15	ec:1.1.3.46	ec:1.14.13.21 5	ec:4.1.1.109
ec:1.13.11.20	ec:1.14.14.106	ec:1.7.3.1	ec:1.14.14.63	ec:1.4.3.10	ec:1.14.14.19	ec:1.1.3.-	ec:1.14.13.21 9	ec:7.1.1.4
ec:1.13.11.25	ec:1.14.14.110	ec:1.7.3.6	ec:1.14.14.67	ec:1.4.3.14	ec:1.14.14.23	ec:1.1.-.-	ec:1.14.13.22 4	
ec:1.13.11.29	ec:1.14.14.114	ec:1.8.3.4	ec:1.14.14.71	ec:1.4.3.20	ec:1.14.14.27	ec:1.2.3.5	ec:1.14.13.22 8	
ec:1.13.11.34	ec:1.14.14.118	ec:1.10.3.1	ec:1.14.14.75	ec:1.4.3.24	ec:1.14.14.32	ec:1.2.3.9	ec:1.14.13.23 2	
ec:1.13.11.38	ec:1.14.14.122	ec:1.10.3.5	ec:1.14.14.79	ec:1.4.9.2	ec:1.14.14.36	ec:1.2.3.-	ec:1.14.13.23 6	
ec:1.13.11.43	ec:1.14.14.126	ec:1.10.3.15	ec:1.14.14.83	ec:1.5.3.5	ec:1.14.14.40	ec:1.3.3.6	ec:1.14.13.24 0	
ec:1.13.11.48	ec:1.14.14.130	ec:1.11.1.6	ec:1.14.14.87	ec:1.5.3.12	ec:1.14.14.44	ec:1.3.3.11	ec:1.14.13.24 4	
ec:1.13.11.52	ec:1.14.14.134	ec:1.13.11.3	ec:1.14.14.91	ec:1.5.3.16	ec:1.14.14.48	ec:1.3.3.15	ec:1.14.13.-	
ec:1.13.11.56	ec:1.14.14.138	ec:1.13.11.8	ec:1.14.14.95	ec:1.5.3.21	ec:1.14.14.52	ec:1.3.99.28	ec:1.14.14.8	
ec:1.13.11.60	ec:1.14.14.143	ec:1.13.11.1 2	ec:1.14.14.99	ec:1.5.99.6	ec:1.14.14.56	ec:1.4.3.1	ec:1.14.14.12	
ec:1.13.11.64	ec:1.14.14.147	ec:1.13.11.1 7	ec:1.14.14.10 3	ec:1.6.3.4	ec:1.14.14.60	ec:1.4.3.5	ec:1.14.14.16	
ec:1.13.11.68	ec:1.14.14.151	ec:1.13.11.2 2	ec:1.14.14.10 7	ec:1.7.3.2	ec:1.14.14.64	ec:1.4.3.11	ec:1.14.14.20	
ec:1.13.11.72	ec:1.14.14.155	ec:1.13.11.2 6	ec:1.14.14.11 1	ec:1.8.3.1	ec:1.14.14.68	ec:1.4.3.15	ec:1.14.14.24	
ec:1.13.11.76	ec:1.14.14.159	ec:1.13.11.3 0	ec:1.14.14.11 5	ec:1.8.3.5	ec:1.14.14.72	ec:1.4.3.21	ec:1.14.14.29	
ec:1.13.11.80	ec:1.14.14.163	ec:1.13.11.3 5	ec:1.14.14.11 9	ec:1.10.3.2	ec:1.14.14.76	ec:1.4.3.25	ec:1.14.14.33	
ec:1.13.11.84	ec:1.14.14.167	ec:1.13.11.3 9	ec:1.14.14.12 3	ec:1.10.3.6	ec:1.14.14.80	ec:1.5.3.1	ec:1.14.14.37	
ec:1.13.11.89	ec:1.14.14.171	ec:1.13.11.4 5	ec:1.14.14.12 7	ec:1.10.3.16	ec:1.14.14.84	ec:1.5.3.6	ec:1.14.14.41	
ec:1.13.11.-	ec:1.14.14.175	ec:1.13.11.4 9	ec:1.14.14.13 1	ec:1.11.1.21	ec:1.14.14.88	ec:1.5.3.13	ec:1.14.14.45	
ec:1.13.12.4	ec:1.14.15.1	ec:1.13.11.5 3	ec:1.14.14.13 5	ec:1.13.11.4	ec:1.14.14.92	ec:1.5.3.17	ec:1.14.14.49	
ec:1.13.12.8	ec:1.14.15.6	ec:1.13.11.5 7	ec:1.14.14.13 9	ec:1.13.11.9	ec:1.14.14.96	ec:1.5.3.22	ec:1.14.14.53	
ec:1.13.12.16	ec:1.14.15.10	ec:1.13.11.6 1	ec:1.14.14.14 4	ec:1.13.11.1 4	ec:1.14.14.10 0	ec:1.6.3.1	ec:1.14.14.57	
ec:1.13.12.20	ec:1.14.15.16	ec:1.13.11.6 5	ec:1.14.14.14 8	ec:1.13.11.1 8	ec:1.14.14.10 4	ec:1.6.3.5	ec:1.14.14.61	
ec:1.13.12.24	ec:1.14.15.20	ec:1.13.11.6 9	ec:1.14.14.15 2	ec:1.13.11.2 3	ec:1.14.14.10 8	ec:1.7.3.3	ec:1.14.14.65	
ec:1.13.-.-	ec:1.14.15.24	ec:1.13.11.7 3	ec:1.14.14.15 6	ec:1.13.11.2 7	ec:1.14.14.11 2	ec:1.8.3.2	ec:1.14.14.69	

Table S3. (continued)

ec:1.14.11.4	ec:1.14.15.28	ec:1.13.11.7 7	ec:1.14.14.16 0	ec:1.13.11.3 1	ec:1.14.14.11 6	ec:1.8.3.6	ec:1.14.14.73	
ec:1.14.11.9	ec:1.14.15.32	ec:1.13.11.8 1	ec:1.14.14.16 4	ec:1.13.11.3 6	ec:1.14.14.12 0	ec:1.10.3.3	ec:1.14.14.77	
ec:1.14.11.13	ec:1.14.15.36	ec:1.13.11.8 5	ec:1.14.14.16 8	ec:1.13.11.4 0	ec:1.14.14.12 4	ec:1.10.3.9	ec:1.14.14.81	
ec:1.14.11.18	ec:1.14.15.-	ec:1.13.11.9 0	ec:1.14.14.17 2	ec:1.13.11.4 6	ec:1.14.14.12 8	ec:1.10.3.17	ec:1.14.14.85	
ec:1.14.11.25	ec:1.14.16.5	ec:1.13.12.1	ec:1.14.14.17 6	ec:1.13.11.5 0	ec:1.14.14.13 2	ec:1.13.11.1	ec:1.14.14.89	
ec:1.14.11.31	ec:1.14.17.1	ec:1.13.12.5	ec:1.14.15.3	ec:1.13.11.5 4	ec:1.14.14.13 6	ec:1.13.11.5	ec:1.14.14.93	
ec:1.14.11.37	ec:1.14.18.2	ec:1.13.12.9	ec:1.14.15.7	ec:1.13.11.5 8	ec:1.14.14.14 1	ec:1.13.11.1 0	ec:1.14.14.97	
ec:1.14.11.41	ec:1.14.18.6	ec:1.13.12.1 7	ec:1.14.15.11	ec:1.13.11.6 2	ec:1.14.14.14 5	ec:1.13.11.1 5	ec:1.14.14.10 1	
ec:1.14.11.45	ec:1.14.18.11	ec:1.13.12.2 1	ec:1.14.15.17	ec:1.13.11.6 6	ec:1.14.14.14 9	ec:1.13.11.1 9	ec:1.14.14.10 5	
ec:1.14.11.51	ec:1.14.19.2	ec:1.13.12.-	ec:1.14.15.21	ec:1.13.11.7 0	ec:1.14.14.15 3	ec:1.13.11.2 4	ec:1.14.14.10 9	
ec:1.14.11.57	ec:1.14.19.9	ec:1.14.11.1	ec:1.14.15.25	ec:1.13.11.7 4	ec:1.14.14.15 7	ec:1.13.11.2 8	ec:1.14.14.11 3	
ec:1.14.11.61	ec:1.14.19.13	ec:1.14.11.6	ec:1.14.15.29	ec:1.13.11.7 8	ec:1.14.14.16 1	ec:1.13.11.3	ec:1.14.14.11 7	
ec:1.14.11.65	ec:1.14.19.19	ec:1.14.11.1 0	ec:1.14.15.33	ec:1.13.11.8 2	ec:1.14.14.16 5	ec:1.13.11.3 7	ec:1.14.14.12 1	
ec:1.14.11.69	ec:1.14.19.26	ec:1.14.11.1 5	ec:1.14.15.37	ec:1.13.11.8 6	ec:1.14.14.16 9	ec:1.13.11.4 1	ec:1.14.14.12 5	
ec:1.14.11.73	ec:1.14.19.39	ec:1.14.11.2 0	ec:1.14.16.1	ec:1.13.11.9 1	ec:1.14.14.17 3	ec:1.13.11.4 7	ec:1.14.14.12 9	
ec:1.14.11.77	ec:1.14.19.49	ec:1.14.11.2 6	ec:1.14.16.6	ec:1.13.12.2	ec:1.14.14.17 7	ec:1.13.11.5 1	ec:1.14.14.13 3	
ec:1.14.12.7	ec:1.14.19.53	ec:1.14.11.3 2	ec:1.14.17.3	ec:1.13.12.6	ec:1.14.15.4	ec:1.13.11.5 5	ec:1.14.14.13 7	
ec:1.14.12.11	ec:1.14.19.57	ec:1.14.11.3 8	ec:1.14.18.3	ec:1.13.12.1 3	ec:1.14.15.8	ec:1.13.11.5 9	ec:1.14.14.14 2	
ec:1.14.12.15	ec:1.14.19.61	ec:1.14.11.4 2	ec:1.14.18.7	ec:1.13.12.1 8	ec:1.14.15.13	ec:1.13.11.6 3	ec:1.14.14.14 6	
ec:1.14.12.19	ec:1.14.19.65	ec:1.14.11.4 6	ec:1.14.18.12	ec:1.13.12.2 2	ec:1.14.15.18	ec:1.13.11.6 7	ec:1.14.14.15 0	
ec:1.14.12.25	ec:1.14.19.69	ec:1.14.11.5 2	ec:1.14.19.3	ec:1.13.99.1	ec:1.14.15.22	ec:1.13.11.7 1	ec:1.14.14.15 4	
ec:1.14.13.2	ec:1.14.19.73	ec:1.14.11.5 8	ec:1.14.19.10	ec:1.14.11.2	ec:1.14.15.26	ec:1.13.11.7 5	ec:1.14.14.15 8	
ec:1.14.13.7	ec:1.14.19.77	ec:1.14.11.6 2	ec:1.14.19.15	ec:1.14.11.7	ec:1.14.15.30	ec:1.13.11.7 9	ec:1.14.14.16 2	
ec:1.14.13.14	ec:1.14.20.3	ec:1.14.11.6 6	ec:1.14.19.20	ec:1.14.11.1 1	ec:1.14.15.34	ec:1.13.11.8 3	ec:1.14.14.16 6	
ec:1.14.13.20	ec:1.14.20.7	ec:1.14.11.7 0	ec:1.14.19.29	ec:1.14.11.1 6	ec:1.14.15.38	ec:1.13.11.8 8	ec:1.14.14.17 0	
ec:1.14.13.25	ec:1.14.20.11	ec:1.14.11.7 4	ec:1.14.19.41	ec:1.14.11.2 1	ec:1.14.16.2	ec:1.13.11.9 2	ec:1.14.14.17 4	
ec:1.14.13.32	ec:1.14.20.15	ec:1.14.11.-	ec:1.14.19.50	ec:1.14.11.2 7	ec:1.14.16.7	ec:1.13.12.3	ec:1.14.14.-	
ec:1.14.13.38	ec:1.14.99.2	ec:1.14.12.8	ec:1.14.19.54	ec:1.14.11.3 5	ec:1.14.17.4	ec:1.13.12.7	ec:1.14.15.5	
ec:1.14.13.44	ec:1.14.99.14	ec:1.14.12.1 2	ec:1.14.19.58	ec:1.14.11.3 9	ec:1.14.18.4	ec:1.13.12.1 5	ec:1.14.15.9	
ec:1.14.13.54	ec:1.14.99.22	ec:1.14.12.1 6	ec:1.14.19.62	ec:1.14.11.4 3	ec:1.14.18.9	ec:1.13.12.1 9	ec:1.14.15.15	
ec:1.14.13.62	ec:1.14.99.29	ec:1.14.12.2 2	ec:1.14.19.66	ec:1.14.11.4 8	ec:1.14.18.-	ec:1.13.12.2 3	ec:1.14.15.19	
ec:1.14.13.69	ec:1.14.99.39	ec:1.14.12.2 6	ec:1.14.19.70	ec:1.14.11.5 5	ec:1.14.19.6	ec:1.13.99.3	ec:1.14.15.23	
ec:1.14.13.84	ec:1.14.99.48	ec:1.14.13.4	ec:1.14.19.74	ec:1.14.11.5 9	ec:1.14.19.11	ec:1.14.11.3	ec:1.14.15.27	
ec:1.14.13.10 7	ec:1.14.99.53	ec:1.14.13.8	ec:1.14.19.78	ec:1.14.11.6 3	ec:1.14.19.17	ec:1.14.11.8	ec:1.14.15.31	

Table S3. (continued)

RC00714	1	RC00714	O-Hydrolysis
RC00805	1	RC00805	De/hydrogenation
RC01114	1	RC01114	Anthocyanin/din di/hydroxylation
RC01553	1	RC01553	Anthocyanin/din de/hydrogenation
RC01632	1	RC01632	Di/hydroxylation
RC01837	1	RC01837	Di/hydroxylation
RC00046	1	RC00046	Di/hydroxylation
RC01587	1	RC01587	C-Hydrolysis
RC00234	1	RC00234	De/hydrogenation
RC00235	1	RC00235	De/hydrogenation
RC00657	1	RC00657	De/hydrogenation
RC00717	1	RC00717	De/hydrogenation
RC01235	1	RC01235	De/hydrogenation
RC00392	1	RC00392	De/methylation
RC00230	1	RC00230	Di/hydroxylation
RC00235	1	RC00235	Di/hydroxylation
RC00490	1	RC00490	Di/hydroxylation
RC01552	1	RC01552	Di/hydroxylation
RC00049	1	RC00049	O-Hydrolysis
RC00171	1	RC00171	O-Hydrolysis
RC00718	1	RC00718	Ring cleavage
RC02001	1	RC02001	Ring cleavage
RC02019	1	RC02019	Ring cleavage
RC02726	1	RC02726	Ring cleavage
<u>RC02046</u>	0.9	RC01632	Di/hydroxylation
<u>RC00490</u>	0.9	RC00046	Di/hydroxylation
<u>RC00236</u>	0.9	RC00046	Di/hydroxylation
<u>RC00613</u>	0.9	RC00230	Di/hydroxylation
<u>RC00046</u>	0.9	RC00490	Di/hydroxylation
<u>RC02573</u>	0.9	RC00490	Di/hydroxylation
<u>RC00589</u>	0.9	RC00490	Di/hydroxylation
<u>RC01856</u>	0.9	RC00490	Di/hydroxylation
<u>RC02016</u>	0.9	RC00490	Di/hydroxylation
<u>RC00613</u>	0.9	RC01552	Di/hydroxylation
<u>RC00274</u>	0.9	RC01552	Di/hydroxylation
<u>RC00432</u>	0.9	RC01587	C-Hydrolysis
<u>RC02019</u>	0.9	RC02001	Ring cleavage

Table S4.

Master RClass similarity table with a cutoff value of 0.8

<u>RC02001</u>	0.9	RC02019	Ring cleavage
<u>RC01053</u>	0.877	RC01632	Di/hydroxylation
<u>RC01414</u>	0.877	RC01632	Di/hydroxylation
<u>RC02573</u>	0.867	RC00046	Di/hydroxylation
<u>RC00589</u>	0.867	RC00046	Di/hydroxylation
<u>RC01856</u>	0.867	RC00046	Di/hydroxylation
<u>RC02016</u>	0.867	RC00046	Di/hydroxylation
<u>RC03365</u>	0.867	RC00046	Di/hydroxylation
<u>RC01860</u>	0.867	RC00046	Di/hydroxylation
<u>RC00661</u>	0.867	RC00230	Di/hydroxylation
<u>RC00704</u>	0.867	RC00230	Di/hydroxylation
<u>RC01993</u>	0.867	RC00230	Di/hydroxylation
<u>RC00524</u>	0.867	RC00230	Di/hydroxylation
<u>RC02085</u>	0.867	RC00230	Di/hydroxylation
<u>RC01552</u>	0.867	RC00230	Di/hydroxylation
<u>RC02734</u>	0.867	RC00230	Di/hydroxylation
<u>RC03493</u>	0.867	RC00230	Di/hydroxylation
<u>RC02043</u>	0.867	RC00230	Di/hydroxylation
<u>RC00661</u>	0.867	RC01552	Di/hydroxylation
<u>RC00277</u>	0.867	RC01552	Di/hydroxylation
<u>RC01993</u>	0.867	RC01552	Di/hydroxylation
<u>RC00524</u>	0.867	RC01552	Di/hydroxylation
<u>RC02085</u>	0.867	RC01552	Di/hydroxylation
<u>RC00230</u>	0.867	RC01552	Di/hydroxylation
<u>RC02734</u>	0.867	RC01552	Di/hydroxylation
<u>RC02043</u>	0.867	RC01552	Di/hydroxylation
<u>RC03497</u>	0.867	RC01552	Di/hydroxylation
<u>RC02955</u>	0.867	RC02726	Ring cleavage
<u>RC02969</u>	0.867	RC02726	Ring cleavage
<u>RC03232</u>	0.86	RC01632	Di/hydroxylation
<u>RC00052</u>	0.854	RC01235	De/hydrogenation
<u>RC02116</u>	0.854	RC01235	De/hydrogenation
<u>RC00144</u>	0.833	RC00234	De/hydrogenation
<u>RC00273</u>	0.833	RC00234	De/hydrogenation
<u>RC00182</u>	0.833	RC00235	De/hydrogenation
<u>RC00273</u>	0.833	RC00235	De/hydrogenation
<u>RC02358</u>	0.833	RC01235	De/hydrogenation
<u>RC01836</u>	0.833	RC00392	De/methylation
<u>RC01916</u>	0.833	RC00392	De/methylation

Table S4. (continued)

RC01558	0.833	RC00392	De/methylation
RC00505	0.833	RC00046	Di/hydroxylation
RC00716	0.833	RC00046	Di/hydroxylation
RC01316	0.833	RC00046	Di/hydroxylation
RC00412	0.833	RC00046	Di/hydroxylation
RC01801	0.833	RC00046	Di/hydroxylation
RC00570	0.833	RC00046	Di/hydroxylation
RC01919	0.833	RC00046	Di/hydroxylation
RC00901	0.833	RC00046	Di/hydroxylation
RC01836	0.833	RC00046	Di/hydroxylation
RC00182	0.833	RC00235	Di/hydroxylation
RC00273	0.833	RC00235	Di/hydroxylation
RC01492	0.833	RC01587	C-Hydrolysis
RC00452	0.833	RC00049	O-Hydrolysis
RC01992	0.833	RC00049	O-Hydrolysis
RC01669	0.833	RC00049	O-Hydrolysis
RC01776	0.833	RC00049	O-Hydrolysis
RC01684	0.833	RC00171	O-Hydrolysis
RC02508	0.833	RC00171	O-Hydrolysis
RC02023	0.833	RC00171	O-Hydrolysis
RC02046	0.817	RC01114	Anthocyanin/din di/hydroxylation
RC00466	0.8	RC00714	O-Hydrolysis
RC00451	0.8	RC00714	O-Hydrolysis
RC00746	0.8	RC00714	O-Hydrolysis
RC01248	0.8	RC00714	O-Hydrolysis
RC00049	0.8	RC00714	O-Hydrolysis
RC00077	0.8	RC00714	O-Hydrolysis
RC00350	0.8	RC00714	O-Hydrolysis
RC00233	0.8	RC00714	O-Hydrolysis
RC00878	0.8	RC00714	O-Hydrolysis
RC00530	0.8	RC00714	O-Hydrolysis
RC00231	0.8	RC00714	O-Hydrolysis
RC02123	0.8	RC00392	De/methylation
RC01746	0.8	RC00392	De/methylation
RC02082	0.8	RC00392	De/methylation
RC00466	0.8	RC00392	De/methylation
RC01324	0.8	RC00392	De/methylation
RC01895	0.8	RC00392	De/methylation
RC01372	0.8	RC00392	De/methylation

Table S4. (continued)

RC03192	0.8	RC00392	De/methylation
RC00171	0.8	RC00392	De/methylation
RC00878	0.8	RC00392	De/methylation
RC00128	0.8	RC00392	De/methylation
RC00332	0.8	RC00046	Di/hydroxylation
RC00209	0.8	RC00046	Di/hydroxylation
RC00701	0.8	RC00046	Di/hydroxylation
RC03141	0.8	RC00046	Di/hydroxylation
RC00432	0.8	RC00046	Di/hydroxylation
RC01590	0.8	RC00046	Di/hydroxylation
RC01921	0.8	RC00046	Di/hydroxylation
RC00390	0.8	RC00046	Di/hydroxylation
RC01838	0.8	RC00046	Di/hydroxylation
RC02112	0.8	RC00046	Di/hydroxylation
RC01199	0.8	RC00046	Di/hydroxylation
RC03483	0.8	RC00046	Di/hydroxylation
RC01913	0.8	RC00046	Di/hydroxylation
RC00949	0.8	RC00046	Di/hydroxylation
RC03265	0.8	RC00230	Di/hydroxylation
RC00478	0.8	RC00230	Di/hydroxylation
RC01218	0.8	RC00230	Di/hydroxylation
RC01963	0.8	RC00230	Di/hydroxylation
RC00723	0.8	RC00230	Di/hydroxylation
RC00277	0.8	RC00230	Di/hydroxylation
RC02241	0.8	RC00230	Di/hydroxylation
RC01203	0.8	RC00230	Di/hydroxylation
RC01480	0.8	RC00230	Di/hydroxylation
RC02731	0.8	RC00230	Di/hydroxylation
RC01592	0.8	RC00230	Di/hydroxylation
RC00274	0.8	RC00230	Di/hydroxylation
RC03497	0.8	RC00230	Di/hydroxylation
RC00364	0.8	RC00490	Di/hydroxylation
RC00391	0.8	RC00490	Di/hydroxylation
RC00236	0.8	RC00490	Di/hydroxylation
RC03365	0.8	RC00490	Di/hydroxylation
RC01860	0.8	RC00490	Di/hydroxylation
RC01637	0.8	RC00490	Di/hydroxylation
RC00478	0.8	RC01552	Di/hydroxylation
RC01218	0.8	RC01552	Di/hydroxylation

Table S4. (continued)

RC01963	0.8	RC01552	Di/hydroxylation
RC00723	0.8	RC01552	Di/hydroxylation
RC00704	0.8	RC01552	Di/hydroxylation
RC02241	0.8	RC01552	Di/hydroxylation
RC01203	0.8	RC01552	Di/hydroxylation
RC01480	0.8	RC01552	Di/hydroxylation
RC03493	0.8	RC01552	Di/hydroxylation
RC02731	0.8	RC01552	Di/hydroxylation
RC01592	0.8	RC01552	Di/hydroxylation
RC01662	0.8	RC01587	C-Hydrolysis
RC01869	0.8	RC01587	C-Hydrolysis
RC03346	0.8	RC01587	C-Hydrolysis
RC02379	0.8	RC01587	C-Hydrolysis
RC00801	0.8	RC01587	C-Hydrolysis
RC00569	0.8	RC01587	C-Hydrolysis
RC00236	0.8	RC01587	C-Hydrolysis
RC01363	0.8	RC01587	C-Hydrolysis
RC00466	0.8	RC00049	O-Hydrolysis
RC00451	0.8	RC00049	O-Hydrolysis
RC00746	0.8	RC00049	O-Hydrolysis
RC01248	0.8	RC00049	O-Hydrolysis
RC00059	0.8	RC00049	O-Hydrolysis
RC00397	0.8	RC00049	O-Hydrolysis
RC00262	0.8	RC00049	O-Hydrolysis
RC00028	0.8	RC00049	O-Hydrolysis
RC02398	0.8	RC00049	O-Hydrolysis
RC00171	0.8	RC00049	O-Hydrolysis
RC00529	0.8	RC00049	O-Hydrolysis
RC00077	0.8	RC00049	O-Hydrolysis
RC00350	0.8	RC00049	O-Hydrolysis
RC00714	0.8	RC00049	O-Hydrolysis
RC00530	0.8	RC00049	O-Hydrolysis
RC00231	0.8	RC00049	O-Hydrolysis
RC00392	0.8	RC00171	O-Hydrolysis
RC01372	0.8	RC00171	O-Hydrolysis
RC03192	0.8	RC00171	O-Hydrolysis
RC00059	0.8	RC00171	O-Hydrolysis
RC00397	0.8	RC00171	O-Hydrolysis
RC00262	0.8	RC00171	O-Hydrolysis

Table S4. (continued)

<u>RC00049</u>	0.8	RC00171	O-Hydrolysis
<u>RC00028</u>	0.8	RC00171	O-Hydrolysis
<u>RC02398</u>	0.8	RC00171	O-Hydrolysis
<u>RC00529</u>	0.8	RC00171	O-Hydrolysis
<u>RC00878</u>	0.8	RC00171	O-Hydrolysis
<u>RC00128</u>	0.8	RC00171	O-Hydrolysis

Table S4. (continued)

Input: Enzyme

Below code can take an enzyme in the form of EC number and outputs a list of predictions.

Example input: enzyme = 'ec:2.3.1.74'

```
[num, txt, raw] = xlsread('reclass_sim_list_refined_v2_20211216.xlsx');
rc_sims = raw;

%read the KEGG API url to get the reclass(es) for the input enzyme
url1 = 'http://rest.kegg.jp/link/rc/';
url2 = enzyme;
url_api = strcat(url1,url2);
str_read = urlread(url_api);
index = strfind(str_read,'rc:');

%in a for loop, get all the reclasses based on the index and write them into
%a new vector called reclasses
count=1;
for j=1:length(index)
    new_index_init = index(j)+3;
    new_index_end = new_index_init + 7;
    reclasses(count) = string(str_read(new_index_init:new_index_end));
    count = count + 1;
end

reclasses = cellstr(transpose(unique(strtrim(reclasses))));
%now we will check if the reclass(es) belong to any of our reaction
%categories

count=1;
for i=1:length(reclasses)
    i
    idx = find(contains(rc_sims(:,1), reclasses{i}));
    if length(idx) ~= 0
        for j=1:length(idx)
            rxn = rc_sims{idx(j), 4};
            score = rc_sims{idx(j), 2};
            key_reclass = rc_sims{idx(j),3};
            reclass_cpds = reclass_to_cpds(key_reclass);
            cpds = refine_cpds(reclass_cpds);
            results(count).reaction = rxn;
            results(count).score = score;
            results(count).compounds = cpds;
            count = count + 1;
        end
    end
end
end
T = struct2table(results);
sortedT = sortrows(T, 'score', 'descend');
sortedS = table2struct(sortedT);
```

Input: Organism

Below code can take an organism in the form of KEGG organism ID and outputs a list of predictions. Example input: mic = 'fpla'

```
[num, txt, raw] = xlsread('rclass_sim_list_refined_v2_cutoff08.xlsx');
rc_sims = raw;
keys = txt;

%retrieve the enzyme list for the given microbe
url_mic = mic;
url_link = 'http://rest.kegg.jp/link/ec/';
url_all = strcat(url_link,url_mic);
ecs_read = urlread(url_all);
index_ec = strfind(ecs_read,'ec:');
index_mic = strfind(ecs_read, mic);

count=1;
for j=1:length(index_ec)-1
    new_index_init = index_ec(j);
    new_index_end = index_mic(j+1)-1;
    ecs(count) = string(ecs_read(new_index_init:new_index_end));
    count = count + 1;
end

%write the final enzyme
ecs(count) = string(ecs_read(index_ec(length(index_ec)):end));

% new part: remove oxygen containing enzymes
[num, txt, raw] = xlsread('o2_enzymes.xlsx');
o2_ecs = strtrim(txt);

for i=1:length(ecs)
    if ismember(ecs(i),o2_ecs) == 1
        ecs(i) = [];
    end
end
ecs = transpose(unique(strtrim(ecs)));

count=1;
for i=1:length(ecs)
    enzyme = strtrim(ecs(i));

    %read the KEGG API url to get the rclass(es) for the input enzyme
    url1 = 'http://rest.kegg.jp/link/rc/';
    url2 = enzyme;
    url_api = strcat(url1,url2);
    str_read = urlread(url_api);
    index = strfind(str_read,'rc:');

    %in a for loop, get all the rclasses based on the index and write them into
    %a new vector called rclasses
```

```

if isempty(index) == 0
    for j=1:length(index)
        new_index_init = index(j)+3;
        new_index_end = new_index_init + 7;
        rclasses(count) = string(str_read(new_index_init:new_index_end));
        ec_rc_map(count,1) = enzyme;
        ec_rc_map(count,2) = strtrim(rclasses(count));
        count = count + 1;
    end
end
end

rclasses = cellstr(transpose(unique(strtrim(rclasses))));
%now we will check if the rclass(es) belong to any of our reaction
%categories
struct_count = 1;
for i=1:length(rclasses)

    idx1 = find(contains(keys(:,1), rclasses{i}));
    idx2 = find(contains(ec_rc_map(:,2), rclasses{i}));
    for j=1:length(idx1)
        rxn = rc_sims{idx1(j), 4};
        score = rc_sims{idx1(j), 2};
        key_rclass = rc_sims{idx1(j),3};
        rclass_cpds = rclass_to_cpds(key_rclass);
        cpds = refine_cpds(rclass_cpds);
        results(struct_count).enzyme = ec_rc_map(idx2,1);
        results(struct_count).rclass = ec_rc_map(idx2,2);
        results(struct_count).reaction = rxn;
        results(struct_count).score = score;
        results(struct_count).compounds = cpds;
        struct_count = struct_count + 1;
    end
end

T = struct2table(results);
sortedT = sortrows(T, 'reaction', 'descend');
sortedS = table2struct(sortedT);

```

Retrieving the compounds from a given RClass

%this function takes the rclass and brings the compounds that are
%associated with it through reactions database

```
function[cpds] = rclass_to_cpds(rclass)

%first get reactions from rclass
rc = rclass;
url1 = 'http://rest.kegg.jp/link/rn';
url_api = strcat(url1,rc);
rxn_read = urlread(url_api);

count=1;
index = strfind(rxn_read,'rn:');
for j=1:length(index)
    new_index_init = index(j)+3;
    new_index_end = new_index_init + 6;
    rxns(count) = string(rxn_read(new_index_init:new_index_end));
    count = count + 1;
end

%now use reactions to get compounds
url2 = 'http://rest.kegg.jp/link/cpd/';
count=1;
for i=1:length(rxns)
    rxn_id = rxns{i};
    url_api2 = strcat(url2, rxn_id);
    cpd_read = urlread(url_api2);
    index_cpd = strfind(cpd_read,'cpd:');
    for j=1:length(index_cpd)
        new_index_init = index_cpd(j)+4;
        new_index_end = new_index_init+6;
        cpds(count) = string(cpd_read(new_index_init:new_index_end));
        count=count+1;
    end
end

cpds = unique(transpose(strtrim(cpds)));
end
```

Refining the compounds to obtain only flavonoids and their first derivatives

%this function takes a vector of compounds and checks if the compound is a
%flavonoid by comparing it to our initial flavonoid compounds database

```
function[flavs] = refine_cpds(cpds)
[num, txt, raw] = xlsread('flavs_list2_20211220.xlsx');
total_flav_list = txt;
Lia = ismember(cpds, total_flav_list);

count=1;
for i=1:length(Lia)
    if Lia(i) == 1
        flavs{count} = cpds{i};
        count=count+1;
    end
end
```

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Reviewer #1 (Public Review):

Summary:

Flavonoids are abundant in plant-based foods. They have been widely recognized for their health-promoting properties. There is increasing evidence that the effects of dietary flavonoids depend on their metabolism by gut bacteria, which can enhance, reduce or otherwise alter the flavonoids' bioactivities. On the other hand, little is known regarding the enzymes and species that can utilize flavonoids as metabolic substrates.

In the current manuscript, the authors analyzed the possibility to predict the degradation of flavonoids that we take up with our food by gut bacteria. In contrast to plants, bacteria do not contain obvious degradation enzymes.

Strengths:

To predict such enzymes with a broad substrate specificity (enzyme promiscuity) the authors optimized/modified a bioinformatic tool to predict whether a gut bacterial enzyme could catalyze a flavonoid reaction based on the chemical reaction similarity of the enzyme's native reaction and known flavonoid reactions in plants.

They predicted such enzyme activities in genomes of bacteria that had been shown to occur in the human gut. Then, they cultivated selected bacteria with the predicted enzymatic activities and in fact showed, that they can degrade parts of these flavonoids. Together with the bioinformatic and mass spectrometry they identified a metabolization pathway of the flavonoid tilianin that spanned multiple species, i.e., *Bifidobacterium longum* subsp. *animalis*, *Blautia coccooides*, and *Flavonifractor plautii*. Lastly, the authors showed that tilianin metabolites exhibit protective effects against H₂O₂ through reactive oxygen species scavenging activity and thus, improve viability of a neuronal cell line, while the parent compound, tilianin, was ineffective. This protective effect might be due to gut microbiota-dependent physiological effects of dietary flavonoids.

Weaknesses:

1. To confirm the bioinformatic-based predictions the authors used in vitro culture experiments and LC-MS experiments. Although these in vitro experiments clearly add value to the bioinformatic prediction, they fall short of providing firm evidence for the predictions because they do not show whether the predicted enzymes really catalyze the predicted reactions. In theory, there could be other enzymes not identified bioinformatically that catalyze the reactions.
2. It is not clear how the authors selected the bacterial species. Did they analyze meta genome sequences or hundreds of genomes of gut bacteria? Did they analyze bacteria isolated from the gut or rather type strains? What about other bacterial species in the gut? Do they also encode relevant enzymes? If yes, how many do? This needs to be clarified.
3. The reported data on *E. coli* is difficult to understand. Has *E. coli* a different degradation pathway leading to the observed disappearance of tilianins?

Reviewer #2 (Public Review):

The manuscript deals with an interesting topic in metabolism: the so-called underground metabolism enabled by enzymes with broad substrate specificity. This is mainly relevant in secondary metabolisms. The authors deal, in particular, with the conversion of flavonoids, which have health-promoting effects. They present an algorithm for predicting the moonlight activities of enzymes, which must be given as inputs. Moreover, the authors performed experiments on the antioxidant activities of the flavonoids under study.

My focus was on the bioinformatics part. Overall, the bioinformatics part is not a major scientific achievement in my eyes, or it is too poorly described to see its merits. There may be

difficulties understanding the presented algorithm.

Comments:

The prediction algorithm should be explained much better. Although the manuscript is quite long, it does not describe the approaches sufficiently well. It is quite hard to read.

As far as I can see, the method was only tested with a small sample of different flavonoid substances.

Major comments

(1) I see the following contradiction. Line 18/19: "As flavonoids are not natural substrates of gut bacterial enzymes" and lines 76/77: "commensal gut microorganisms do not have specialized enzymes that utilize flavonoids as their native substrates" versus lines 72-74: "flavonoids ..., which makes them available to be metabolized". How can they be metabolized given what is said in the first two phrases?

(2) It should be explained better what is meant by "reaction class" (e.g. in lines 97 and 99). Is this the same as the EC number (in the Enzyme Catalogue)? The term "reaction class" is indeed used in the KEGG database. On the webpage

<https://www.genome.jp/brite/br08204>

it seems indeed as if the terms "reaction class" and EC number are somehow equivalent.

However, the term "RClass RC00392" in line 557 of the manuscript points to a difference in meaning.

(3) The prediction algorithm should be explained much better. For example, in the Figure showing the workflow, it is shown that an EC number should be given as input. However, if we search for enzymes which could potentially degrade a given flavonoid, we may not know any suitable EC number. Line 122: "To match a given enzyme with its non-native polyphenolic substrates..." However, where can we take the enzyme name/EC number from? Moreover, given that it is assumed that the reaction is performed by underground metabolism, should the enzyme given as input come from another organism, for example, a plant?

(4) Lines 521-523: Our prediction tool can take either a single enzyme in the form of Enzyme Commission (EC) number (e.g. "ec:2.1.1.75"), or a KEGG organism-identifier (e.g. "cpv") or a consortium, a list of different organism-identifiers, as input." I do not understand the wording "or a consortium". According to the Figure showing the workflow, it should read "and a consortium".

(5) In the Materials and Methods section, the KEGG PATHWAY database is mentioned. This comes somewhat out of the blue. What is the connection to the "reaction class" concept in KEGG? Or is the PATHWAY database only used for extracting the negative controls?

(6) Line 142,143. "Our analysis shows that RClass-based similarity can predict the correct reactions for known flavonoid-metabolizing enzymes". How do the authors know that the results are correct? If it is easy to check, then I assume the test whether a given enzyme is able to catalyze reactions with flavonoids can be done manually in KEGG, so that a computer algorithm is unnecessary.

(7) Elaborating on the previous point - I have the impression that the algorithm is a rather simple search routine for finding reactions in the KEGG database that match certain criteria. This might be a helpful tool to save time in comparison to doing the search manually. However, at least the bioinformatics part of the paper is not a major scientific achievement as far as I can see.

(8) It is not sufficiently clear whether the prediction algorithm only works for the example shown in the top figure (tilianin, acacetin etc), which would be quite a restricted application, or for many or even all flavonoids. In line 565, the authors say: "our tabulated 312 unique flavonoids", while in the upper part of the MS, e.g. in lines 26 and 109, only the pathway starting from tilianin is mentioned.

(9) In which programming language was the algorithm implemented?

(10) The connection between the theoretical and experimental parts of the paper is not fully clear. Some of the experiments serve to test the predictions, which is fine. The experiments on free radicals, however, seem to be somewhat unrelated.